

Testing for Unit Roots in Heterogeneous Panels*

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Abstract

This paper proposes unit root tests for dynamic heterogeneous panels based on the mean of individual unit root statistics. When errors in individual Dickey-Fuller regressions are serially uncorrelated, and normally and independently distributed across groups, it is shown that the proposed LM -bar test statistic is distributed as standard normal for large N (number of groups) and finite T (number of time periods). Exact sample mean and variance of the individual LM statistics needed for the implementation of the test are provided. When errors are serially correlated and heterogeneous across groups, the standardized LM -bar statistic is shown to be valid as T and $N \rightarrow \infty$, with $N/T \rightarrow k$, where k is a finite positive constant. It is also shown that the test is consistent under the alternative hypothesis that the fraction of the individual processes that are stationary is non-zero. In the case where the errors in different regressions contain a common time-specific component we show that our proposed test is still valid so long as the individual LM statistics are based on cross-sectionally de-meaned regressions, but it is now required that both N and T should be sufficiently large. Monte Carlo results confirm better finite sample performances of the proposed panel unit root tests as compared with those proposed by Levin and Lin (1993).

JEL Classification: C12, C15, C22, C23.

Key Words: Heterogeneous dynamic panels, Tests of unit roots, LM -bar and t -bar statistics, Finite sample properties.

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1 Introduction

Since the seminal work of Balestra and Nerlove (1966), dynamic models have played an increasingly important role in empirical analysis of panel data in economics. Given the small time dimension of most panels, the emphasis has been put on models with homogeneous dynamics, and until recently little attention has been paid to the analysis of dynamic heterogeneous panels. However, over the past decade a number of important panel data sets covering different industries, regions, or countries over relatively long time spans have become available, the most prominent example of which is the Summers and Heston (1991) data. The availability of this type of “pseudo” panels raises the issue of the plausibility of the dynamic homogeneity assumption that underlies the traditional analysis of panel data models, and poses the problem of how best to analyze them. The inconsistency of pooled estimators in dynamic heterogeneous panel models has been demonstrated by Pesaran and Smith (1995), and Pesaran, Smith and Im (1996). The present paper builds on these works and considers the problem of testing for unit roots in such pseudo panels.

Panel based unit root tests have been advanced by Quah (1992, 1994) and Levin and Lin (1993, *LL* hereafter). The tests proposed by Quah do not accommodate heterogeneity across groups such as individual specific effects and different patterns of residual serial correlations. *LL*’s test is more generally applicable, allows for individual specific effects as well as dynamic heterogeneity across groups, and requires $N/T \rightarrow 0$ as both N (the number of groups) and T (the number of time periods) tend to infinity.

This paper proposes an alternative testing procedure based on averaging individual unit root test statistics for panels. In particular, we propose a test based on the average of the Lagrange multiplier statistics computed for each group in the panel, which we refer to as the *LM*-bar test. Like the *LL* procedure, our proposed test allows for heterogeneity of the dynamics and error variances across groups, but is more generally applicable, in the sense that it only requires that $N/T \rightarrow k$, for any finite positive constant k , instead of the stronger condition, $N/T \rightarrow 0$, needed for the asymptotic validity of the *LL* test. This suggests that the proposed tests may have better small sample properties than the *LL* test in panels when N is large relative to T . We also prove that the proposed test is consistent under a general class of fixed alternatives that allow for a fraction (strictly less than unity) of the individual group regressions to have unit roots.

In the special case where errors in individual Dickey-Fuller regressions are serially uncorrelated, the *LM*-bar statistic is shown to be distributed as standard normal as $N \rightarrow \infty$ for a finite T . Exact sample mean and variance of the individual *LM* statistics needed for the implementation of the test are also provided.

Finally, in the case where the errors in different regressions contain a common time-specific component we show that our proposed test is still valid so long as the individual *LM* statistics are based on cross-sectionally de-measured regressions, but it is now required that both N and T should be sufficiently large. An attractive feature of this test is its robustness to trends and residual serial correlation patterns that are common across the groups.

The finite sample performances of the proposed *LM*-bar test and an alternative *t*-bar test (based on the average of augmented Dickey-Fuller (*ADF*) *t*-statistics) and the *LL* test are examined using Monte Carlo methods. The simulation results clearly demonstrate the

small sample superiority of both the LM -bar and the t -bar tests over the LL test, with the t -bar test performing marginally better than the LM -bar test.

The plan of the paper is as follows. Section 2 sets out the model and provides a brief review of Quah and LL tests. Section 3 sets out the likelihood framework for heterogeneous panels, and derives the test statistics in the case where errors in individual Dickey-Fuller regressions are serially uncorrelated. Section 4 considers a more general case where errors in individual Dickey-Fuller regressions are serially correlated. Section 5 presents the Monte Carlo evidence on the finite sample performance of the unit root tests. Section 6 provides some concluding remarks. Mathematical proofs are provided in an Appendix.

2 Existing Tests for Unit Roots in Panels

In this section, we introduce the model that underlies testing for unit roots in panels, and briefly review two existing approaches advanced by Quah (1992, 1994) and LL (1993).

Consider a sample of N cross sections (industries, regions or countries) observed over T time periods. We suppose that the stochastic process, y_{it} , is generated by the first-order autoregressive process:

$$y_{it} = (1 - \phi_i) \mu_i + \phi_i y_{i,t-1} + \varepsilon_{it}, \quad i = 1, \dots, N; \quad t = 1, \dots, T, \quad (2.1)$$

where initial values, y_{i0} , are given. We are interested in testing the null hypothesis of unit roots $\phi_i = 1$ for all i . (2.1) can be expressed as

$$\Delta y_{it} = \alpha_i + \beta_i y_{i,t-1} + \varepsilon_{it}, \quad (2.2)$$

where $\alpha_i = (1 - \phi_i) \mu_i$, $\beta_i = -(\phi_i - 1)$ and $\Delta y_{it} = y_{it} - y_{i,t-1}$. The null hypothesis of unit roots then becomes

$$H_0 : \beta_i = 0 \text{ for all } i, \quad (2.3)$$

against the alternatives,

$$H_1 : \beta_i < 0, \quad i = 1, 2, \dots, N_1, \quad \beta_i = 0, \quad i = N_1 + 1, N_1 + 2, \dots, N. \quad (2.4)$$

This formulation of the alternative hypothesis allows for β_i to differ across groups, and is more general than the homogeneous alternative hypothesis, namely $\beta_i = \beta < 0$ for all i , which is implicit in the testing approach of Quah and LL to be discussed below. It also allows for some of the individual series to have unit roots under the alternative hypothesis. As we shall see below, when both N and T are sufficiently large the consistency of our proposed panel unit root test is ensured if under the alternative hypothesis the fraction of the individual processes that are stationary is non-zero, namely if $\lim_{N \rightarrow \infty} (N_1/N) = \delta_1$, $0 < \delta_1 \leq 1$.

Quah (1994) considers the following simple dynamic panel,

$$y_{it} = \phi y_{i,t-1} + \varepsilon_{it}, \quad i = 1, \dots, N; \quad t = 1, \dots, T, \quad (2.5)$$

where ε_{it} are independently and identically distributed both across i and t with finite variance, σ^2 . Under additional conditions that $T = \kappa N$ with $\kappa > 0$, and $c = \lim_{T \rightarrow \infty} E(y_{i0} \times$

$T^{-\frac{1}{2}} \sum_{t=1}^T \varepsilon_{it}$) is the same for all i , Quah shows that under the unit root hypothesis, $\phi = 1$, as $N \rightarrow \infty$ and $T \rightarrow \infty$,

$$Q_{NT}(c, \sigma^2) = \sqrt{\frac{N}{2}} T(\hat{\phi}_{NT} - 1 - 2\frac{c}{\sigma^2} T^{-\frac{3}{2}}) \Rightarrow N(0, 1), \quad (2.6)$$

where $\hat{\phi}_{NT}$ is the pooled OLS estimator of ϕ in (2.5) and “ $\Rightarrow$ ” represents the weak convergence in distribution. $Q_{NT}(c, \sigma^2)$ is of limited practical use as it does not allow for the group specific effects and serially correlated and heterogeneous errors. In another paper Quah (1992) considers the more general case where ε_{it} and y_{i0} follow mixing processes, thus allowing for errors in (2.5) to be serially correlated and to have a certain degree of variance heterogeneity.¹

Levin and Lin (1993) provide a more general testing framework and consider the following three models:

$$\Delta y_{it} = \beta y_{i,t-1} + \alpha_{mi} d_{mt} + u_{it}, \quad i = 1, \dots, N; \quad t = 1, \dots, T; \quad m = 1, 2, 3, \quad (2.7)$$

where d_{mt} contains deterministic variables; $d_{1t} = \{1\}$, $d_{2t} = \{1\}$, and $d_{3t} = \{1, t\}$. In specifying u_{it} , LL allow for different dynamics across groups (see (4.2) below) but also common effects (see (4.9) below), and show that under $\beta = 0$, their proposed statistic weakly converges to a standard normal variate, as $T \rightarrow \infty$ and $N \rightarrow \infty$ with $N/T \rightarrow 0$.

3 Unit Root Tests for Heterogeneous Panels with Serially Uncorrelated Errors

In this section we use the log-likelihood procedure to develop an alternative test of the unit root hypothesis in the context of the panel data model, (2.1), where the errors are serially uncorrelated but T is finite. For this purpose we make the following assumption:

Assumption 3.1 ε_{it} , $i = 1, \dots, N$, $t = 1, \dots, T$, in (2.1) are independently and normally distributed random variables with zero means and finite (possibly) heterogeneous variances, σ_i^2 .

In this case the relevant Dickey-Fuller (1979) regressions are given by (2.2) with the following pooled log-likelihood function:

$$\ell_{NT}(\boldsymbol{\beta}, \boldsymbol{\varphi}) = \sum_{i=1}^N \left\{ -\frac{T}{2} \log 2\pi\sigma_i^2 - \frac{1}{2\sigma_i^2} \sum_{t=1}^T (\Delta y_{it} - \alpha_i - \beta_i y_{i,t-1})^2 \right\}, \quad (3.1)$$

where $\boldsymbol{\beta} = (\beta_1, \dots, \beta_N)'$, $\boldsymbol{\varphi}_i = (\alpha_i, \sigma_i^2)'$ and $\boldsymbol{\varphi} = (\boldsymbol{\varphi}_1', \dots, \boldsymbol{\varphi}_N')'$. Using this likelihood framework one could in principle develop alternative panel unit tests based on the average of the log-likelihood ratio (LR), Wald or the Lagrange multiplier (LM) statistics. Here we focus

¹The application of the mixing condition to y_{i0} with respect to i pre-assumes an immutable ordering of the groups, which is unlikely to exist in practice.

on an LM-based test. This is done primarily for mathematical convenience. Since the LM statistic for testing restrictions in linear regression models can be written as TR^2 , where R is a multiple correlation coefficient ($0 \leq R^2 \leq 1$), it therefore follows that all its moments exist for any finite T . This result does not seem to follow readily for the Wald and the LR statistics.

Let LM_{iT} be the LM statistic for testing $\beta_i = 0$ in (2.2), which is given by²

$$LM_{iT} = \frac{T(\Delta \mathbf{y}_i' \mathbf{M}_\tau \mathbf{y}_{i,-1}) (\mathbf{y}_{i,-1}' \mathbf{M}_\tau \mathbf{y}_{i,-1})^{-1} (\mathbf{y}_{i,-1}' \mathbf{M}_\tau \Delta \mathbf{y}_i)}{(\Delta \mathbf{y}_i' \mathbf{M}_\tau \Delta \mathbf{y}_i)} = \frac{T \Delta \mathbf{y}_i' \mathbf{P}_i \Delta \mathbf{y}_i}{\Delta \mathbf{y}_i' \mathbf{M}_\tau \Delta \mathbf{y}_i}, \quad (3.2)$$

where $\boldsymbol{\tau}_T = (1, 1, \dots, 1)'$ is a $T \times 1$ vector of unity, $\mathbf{M}_\tau = \mathbf{I}_T - \boldsymbol{\tau}_T(\boldsymbol{\tau}_T' \boldsymbol{\tau}_T)^{-1} \boldsymbol{\tau}_T'$ and $\mathbf{P}_i = \mathbf{M}_\tau \mathbf{y}_{i,-1} (\mathbf{y}_{i,-1}' \mathbf{M}_\tau \mathbf{y}_{i,-1})^{-1} \mathbf{y}_{i,-1}' \mathbf{M}_\tau$ are $T \times T$ idempotent matrices of rank $T - 1$ and 1, respectively, $\mathbf{y}_{i,-1} = (y_{i0}, y_{i1}, \dots, y_{i,T-1})'$ and $\Delta \mathbf{y}_i = (\Delta y_{i1}, \Delta y_{i2}, \dots, \Delta y_{iT})'$, then define the average of the N individual LM statistics (called the LM -bar statistic) by

$$\overline{LM}_{NT} = \frac{1}{N} \sum_{i=1}^N LM_{iT}. \quad (3.3)$$

Under $\beta_i = 0$, $\mathbf{y}_{i,-1}$ can be written as

$$\mathbf{y}_{i,-1} = y_{i0} \boldsymbol{\tau}_T + \mathbf{s}_{i,-1}, \quad (3.4)$$

where y_{i0} is a given initial value (fixed or random), $\mathbf{s}_{i,-1} = (s_{i0}, s_{i1}, \dots, s_{i,T-1})'$, with $s_{it} = \sum_{j=1}^t \varepsilon_{ij}$. Let

$$\eta_{iT} = \frac{T(\boldsymbol{\varepsilon}_i' \mathbf{M}_\tau \mathbf{s}_{i,-1})^2}{(\mathbf{s}_{i,-1}' \mathbf{M}_\tau \mathbf{s}_{i,-1})(\boldsymbol{\varepsilon}_i' \mathbf{M}_\tau \boldsymbol{\varepsilon}_i)}, \quad (3.5)$$

it is then easily seen that under $\beta_i = 0$, $LM_{iT} = \eta_{iT}$. Notice that η_{iT} , $i = 1, 2, \dots, N$, are identically and independently distributed with a finite mean $E(\eta_T)$ and a finite variance $Var(\eta_T)$ for any T . Also since $\mathbf{s}_{i,-1} = \mathbf{H}\mathbf{L}\boldsymbol{\varepsilon}_i$ where

$$\mathbf{H}_{T \times T} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 1 & 1 & 0 & \cdots & 0 \\ 1 & 1 & 1 & \cdots & 0 \\ & & & \ddots & \\ 1 & 1 & 1 & \cdots & 1 \end{bmatrix}, \quad \mathbf{L}_{T \times T} = \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 & 0 \\ 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 \\ & & & \ddots & & \\ 0 & 0 & 0 & \cdots & 1 & 0 \end{bmatrix}, \quad (3.6)$$

the expression for η_{iT} can also be written equivalently as

$$\eta_{iT} = \frac{T(\mathbf{v}_i' \mathbf{M}_\tau \mathbf{H}\mathbf{L}\mathbf{v}_i)^2}{(\mathbf{v}_i' \mathbf{L}' \mathbf{H}' \mathbf{M}_\tau \mathbf{H}\mathbf{L}\mathbf{v}_i)(\mathbf{v}_i' \mathbf{M}_\tau \mathbf{v}_i)}, \quad \mathbf{v}_i = \frac{\boldsymbol{\varepsilon}_i}{\sigma_i} \sim N(\mathbf{0}, \mathbf{I}_T). \quad (3.7)$$

The asymptotic distribution of the LM -bar statistic in the case of serially uncorrelated errors is summarized in the following theorem.

²The LM unit root test statistic has been considered by Solo (1984) for single time series (i.e. for $N = 1$).

Theorem 3.1 *Consider the standardized LM-bar statistic*

$$\Gamma_{\overline{LM}} = \frac{\sqrt{N} \{ \overline{LM}_{NT} - E(\eta_T) \}}{\sqrt{\text{Var}(\eta_T)}}, \quad (3.8)$$

where $\overline{LM}_{NT}$ is given by (3.3), and $E(\eta_T)$ and $\text{Var}(\eta_T)$ represent the mean and the variance of η_{iT} defined by (3.7). Under Assumption 3.1 and for $T \geq 3$, as $N \rightarrow \infty$, $\Gamma_{\overline{LM}}$ converges to a standard normal variate under the null hypothesis (2.3).

Proof. See Section A.2 of the Appendix.

Remark 3.1 *The case where the underlying DF regressions contain linear trends can be dealt with in a similar manner. For the $\Gamma_{\overline{LM}}$ statistic to be operational, we need to compute $E(\eta_T)$ and $\text{Var}(\eta_T)$. Table 1 provides these moments for different values of T obtained via stochastic simulation using 50,000 replications in the case of models with an intercept only, and models with a linear time trend.*

Remark 3.2 *As shown in Section A.3 of the Appendix, under $\beta_i < 0$, the LM_{iT} statistic in general depends on $(y_{i0} - \mu_i)/\sigma_i$, when T is finite. Consequently, one cannot rule out the possibility that a test based on the LM-bar statistic could be inconsistent. However, our computations based on stochastic simulations suggest that such a possibility is extremely unlikely in practice, unless $T = 3$ for models with intercepts only, and $T \leq 5$ in the case of models with intercepts and linear time trends. We therefore recommend using a one-sided upper-tail version of the LM-bar test, which for most practical applications will be consistent under the alternative hypothesis (2.4), and assuming that $\lim_{N \rightarrow \infty} (N_1/N) = \delta_1$, $0 < \delta_1 \leq 1$.*

Remark 3.3 *When N and T are both finite, the sample distribution of $\overline{LM}_{NT}$ under the null hypothesis (2.3) is non-standard, but does not depend on any nuisance parameters. In this case the exact sample critical values of the $\overline{LM}_{NT}$ statistic can be computed via stochastic simulation. These critical values are computed using 50,000 replications and summarized in Table 3 for different values of N and T , in the case of models with and without linear time trends. The test based on $\Gamma_{\overline{LM}}$ is equivalent to that based on $\overline{LM}_{NT}$ when N is large enough, but in principle could differ for a finite N . However, our computations suggest that the two testing procedures have very similar performance even for relatively small values of N . To see this note that*

$$\Pr(\Gamma_{\overline{LM}} < z_\alpha) = \Pr(\overline{LM}_{NT} < C_T(\alpha)),$$

where z_α is the α percent critical value of the standard normal distribution, and

$$C_T(\alpha) = z_\alpha \sqrt{N^{-1} \text{Var}(\eta_T)} + E(\eta_T).$$

Hence, the tests based on $\Gamma_{\overline{LM}}$ and $\overline{LM}_{NT}$ differ only to the extent that $C_T(\alpha)$ is different from the corresponding exact critical value of the LM-bar test reported in Table 3. Using the

mean and the variance of η_T given in Table 1 for $T = 10$ we have

N	<u>Approximate 5% Critical Values</u>		<u>Exact 5% Critical Values</u>	
	Without trend	With trend	Without trend	With trend
2	4.428	6.001	4.612	6.017
3	4.054	5.601	4.157	5.612
4	3.831	5.361	3.924	5.355
5	3.679	5.196	3.755	5.192
6	3.567	5.074	3.633	5.078
7	3.480	4.980	3.529	4.975
8	3.410	4.904	3.459	4.908
9	3.351	4.841	3.393	4.839
10	3.302	4.788	3.341	4.784

It is clear from these figures that the standardized LM-bar test provides excellent approximations to the exact test even for relatively small values of N .

Remark 3.4 In the more general case where T_i differs across groups, LM_{iT_i} are independently but not identically distributed across i . Since the third moments of LM_{iT_i} also exist, Theorem 3.1 still holds by the Lyapunov central limit theorem for the following statistic:

$$\Gamma_{\overline{LM}} = \frac{\sqrt{N} \left\{ \overline{LM}_{NT} - N^{-1} \sum_{i=1}^N E(\eta_{T_i}) \right\}}{\sqrt{N^{-1} \sum_{i=1}^N \text{Var}(\eta_{T_i})}}. \quad (3.9)$$

4 Unit Root Tests for Heterogeneous Panels with Serially Correlated Errors

In this section we consider the more general case where the errors in (2.1) may be serially correlated, possibly with different serial correlation patterns across groups. Suppose that y_{it} are generated according to following finite-order $AR(p_i + 1)$ processes:

$$y_{it} = \mu_i \phi_i(1) + \sum_{j=1}^{p_i+1} \phi_{ij} y_{i,t-j} + \varepsilon_{it}, \quad i = 1, \dots, N; \quad t = 1, \dots, T, \quad (4.1)$$

which can also be written equivalently as the $ADF(p_i)$ regressions:

$$\Delta y_{it} = \alpha_i + \beta_i y_{i,t-1} + \sum_{j=1}^{p_i} \rho_{ij} \Delta y_{i,t-j} + \varepsilon_{it}, \quad i = 1, \dots, N; \quad t = 1, \dots, T, \quad (4.2)$$

where $\phi_i(1) = 1 - \sum_{j=1}^{p_i+1} \phi_{ij}$, $\alpha_i = \mu_i \phi_i(1)$, $\beta_i = -\phi_i(1)$, and $\rho_{ij} = -\sum_{h=j+1}^{p_i+1} \phi_{ih}$. Writing the regressions for each group in matrix notations we have,

$$\Delta \mathbf{y}_i = \beta_i \mathbf{y}_{i,-1} + \mathbf{Q}_i \boldsymbol{\gamma}_i + \boldsymbol{\varepsilon}_i, \quad (4.3)$$

where $\mathbf{Q}_i = (\boldsymbol{\tau}_T, \Delta \mathbf{y}_{i,-1}, \Delta \mathbf{y}_{i,-2}, \dots, \Delta \mathbf{y}_{i,-p_i})$ and $\boldsymbol{\gamma}_i = (\alpha_i, \rho_{i1}, \rho_{i2}, \dots, \rho_{ip_i})'$.

We also make the following assumptions:

Assumption 4.1 All the roots of $\phi_i(z) = 1 - \sum_{j=1}^{p_i+1} \phi_{ij} z^j = 0$, $i = 1, 2, \dots, N$, satisfy $|z| \geq 1$. Also, when $\beta_i = 0$, all the roots of $\rho_i(z) = 1 - \sum_{j=1}^{p_i} \rho_{ij} z^j = 0$, $i = 1, 2, \dots, N$, fall outside the unit circle.

Assumption 4.2 ε_{it} , $i = 1, 2, \dots, N$, $t = 1, 2, \dots, T$, in (4.2) are independently distributed as normal variates with zero means and finite (possibly) heterogeneous variances, σ_i^2 , and the initial values, $y_{i0}, y_{i,-1}, \dots, y_{i,-p}$, are given (either fixed or stochastic).

Now, the average LM statistic across groups is defined by

$$\overline{LM}_{NT} = \frac{1}{N} \sum_{i=1}^N LM_{iT}(p_i, \boldsymbol{\rho}_i), \quad (4.4)$$

where $\mathbf{p} = (p_1, p_2, \dots, p_N)'$, $\boldsymbol{\rho}_i = (\rho_{i1}, \rho_{i2}, \dots, \rho_{ip_i})'$, $\boldsymbol{\rho} = (\boldsymbol{\rho}'_1, \boldsymbol{\rho}'_2, \dots, \boldsymbol{\rho}'_N)'$ and $LM_{iT}(p_i, \boldsymbol{\rho}_i)$ is the individual LM statistic for testing $\beta_i = 0$ in (4.2) given by

$$LM_{iT}(p_i, \boldsymbol{\rho}_i) = T \frac{(\Delta \mathbf{y}'_i \mathbf{M}_{Q_i} \mathbf{y}_{i,-1}) (\mathbf{y}'_{i,-1} \mathbf{M}_{Q_i} \mathbf{y}_{i,-1})^{-1} (\mathbf{y}'_{i,-1} \mathbf{M}_{Q_i} \Delta \mathbf{y}_i)}{(\Delta \mathbf{y}'_i \mathbf{M}_{Q_i} \Delta \mathbf{y}_i)} = T \frac{(\Delta \mathbf{y}'_i \mathbf{P}_i^* \Delta \mathbf{y}_i)}{(\Delta \mathbf{y}'_i \mathbf{M}_{Q_i} \Delta \mathbf{y}_i)}, \quad (4.5)$$

where

$$\mathbf{M}_{Q_i} = \mathbf{I}_T - \mathbf{Q}_i(\mathbf{Q}'_i \mathbf{Q}_i)^{-1} \mathbf{Q}'_i, \quad \mathbf{P}_i^* = \mathbf{M}_{Q_i} \mathbf{y}_{i,-1} (\mathbf{y}'_{i,-1} \mathbf{M}_{Q_i} \mathbf{y}_{i,-1})^{-1} \mathbf{y}'_{i,-1} \mathbf{M}_{Q_i},$$

and $\mathbf{M}_{Q_i}$ and $\mathbf{P}_i^*$ are $T \times T$ idempotent matrices of rank $T - p_i - 1$ and 1, respectively.

When T is finite the individual LM statistics, $LM_{iT}(p_i, \boldsymbol{\rho}_i)$, will depend on the nuisance parameters, $\boldsymbol{\rho}_i$, even under $\beta_i = 0$. But it turns out that the standardized LM-bar statistic (3.8) will still be valid asymptotically, but it is now required that both N and T be sufficiently large such that $N/T \rightarrow k$, for a finite positive constant k . A more precise statement of this result is given in the following theorem.

Theorem 4.1 Under Assumptions 4.1 - 4.2, and the null hypothesis, $\beta_i = 0$, $i = 1, 2, \dots, N$, in (4.2), the standardized LM-bar statistic, $\Gamma_{\overline{LM}}$, defined by (3.8), weakly converges to a standard normal variate as T and $N \rightarrow \infty$, and $N/T \rightarrow k$, where k is a finite (positive) constant. Under the alternative hypothesis (2.4), and assuming that $\lim_{N \rightarrow \infty} (N_1/N) = \delta_1$, $0 < \delta_1 \leq 1$, the $\Gamma_{\overline{LM}}$ statistic diverges without bound.

Proof. See Section A.4 of the Appendix.

In finite samples, the use of means and variances of $LM_{iT}(p_i, \mathbf{0})$ evaluated under $\beta_i = 0$ is more likely to provide better approximations. This is because $E[LM_{iT}(p_i, \mathbf{0}) | \beta_i = 0]$, for example, makes use of the information contained in p_i while $E[LM_{iT}(0, \mathbf{0}) | \beta_i = 0]$ does not. In view of this we also consider the following modified standardized LM-bar statistic:

$$\Psi_{\overline{LM}} = \frac{\sqrt{N} \left\{ \overline{LM}_{NT} - \frac{1}{N} \sum_{i=1}^N E[LM_{iT}(p_i, \mathbf{0}) | \beta_i = 0] \right\}}{\sqrt{\frac{1}{N} \sum_{i=1}^N Var[LM_{iT}(p_i, \mathbf{0}) | \beta_i = 0]}}. \quad (4.6)$$

Then, under the conditions of Theorem 4.1, it can also be shown that as $T \rightarrow \infty$,

$$\Psi_{\overline{LM}} \Rightarrow N(0, 1). \quad (4.7)$$

The preceding analysis can be readily extended to dynamic panels with intercepts and linear time trends. The values of $E[LM_{iT}(p_i, \mathbf{0}) \mid \beta_i = 0]$ and $Var[LM_{iT}(p_i, \mathbf{0}) \mid \beta_i = 0]$ can be computed via stochastic simulations. Table 1 gives the values of these moments computed using 50,000 replications for different values of T and p , when the underlying ADF(p) regression is estimated with and without a linear time trend.

4.1 Panels with unobserved common time-specific components

There are cases where the disturbances may be correlated across groups. In such circumstances our test procedure, which requires that the observations are generated independently across groups, is no longer applicable. In general, as noted by Quah (1994), modelling cross-sectional dependence is more complicated because unlike the time-series observations individual observations in cross-section display no natural ordering. To allow for the possibility of correlated errors across groups we consider the following model:

$$\Delta y_{it} = \alpha_i + \beta_i y_{i,t-1} + u_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T, \quad (4.8)$$

where we assume that u_{it} is composed of two random components,

$$u_{it} = \theta_t + \varepsilon_{it}, \quad (4.9)$$

a time-specific common effect, θ_t , that allows for a degree of dependency across groups, and an idiosyncratic random effect, ε_{it} , which is independently distributed across groups. To apply our unit root testing procedure, we first need to remove the effect of the common component, θ_t . Subtracting cross-section means from both sides of (4.8), we have³

$$\Delta \tilde{y}_{it} = \tilde{\alpha}_i + \beta_i \tilde{y}_{i,t-1} + \tilde{\xi}_{it}, \quad (4.10)$$

where

$$\tilde{\alpha}_i = \alpha_i - N^{-1} \sum_{j=1}^N \alpha_j, \quad \tilde{y}_{it} = y_{it} - N^{-1} \sum_{j=1}^N y_{jt},$$

and

$$\tilde{\xi}_{it} = \tilde{\varepsilon}_{it} + N^{-1} \sum_{j=1}^N (\beta_i - \beta_j) y_{j,t-1}, \quad \tilde{\varepsilon}_{it} = \varepsilon_{it} - N^{-1} \sum_{j=1}^N \varepsilon_{jt}.$$

The LM -bar testing procedure can now be applied to the above de-meaned regressions to test the null hypothesis that $\beta_i = 0$, $i = 1, 2, \dots, N$. However, the cross-sectional demeaning introduces dependence across the errors, $\tilde{\xi}_{it}$ in the demeaning regressions (4.10), and Theorem 3.1 will not be applicable in the present context. Nevertheless, in the case where the idiosyncratic effects, ε_{it} , are serially uncorrelated, the LM -bar test applied to the de-meaned regressions is asymptotically valid. A formal statement of this result is summarized in the following theorem:

³This de-meaning procedure has been utilized in the panel literature. See Hsiao (1986).

Theorem 4.2 Consider the standardized LM-bar statistic

$$\Gamma_{\widetilde{LM}} = \frac{\sqrt{N} \left\{ \widetilde{LM}_{NT} - E(\eta_T) \right\}}{\sqrt{\text{Var}(\eta_T)}}, \quad (4.11)$$

computed from the demeaned regressions, (4.10) where

$$\widetilde{LM}_{NT} = N^{-1} \sum_{i=1}^N \widetilde{LM}_{iT}, \quad \widetilde{LM}_{iT} = \frac{T (\Delta \tilde{\mathbf{y}}_i' \mathbf{M}_\tau \tilde{\mathbf{y}}_{i,-1})^2}{(\Delta \tilde{\mathbf{y}}_i' \mathbf{M}_\tau \Delta \tilde{\mathbf{y}}_i) (\tilde{\mathbf{y}}_{i,-1}' \mathbf{M}_\tau \tilde{\mathbf{y}}_{i,-1})}, \quad (4.12)$$

$\Delta \tilde{\mathbf{y}}_i = (\Delta \tilde{y}_{i1}, \Delta \tilde{y}_{i2}, \dots, \Delta \tilde{y}_{iT})'$, $\tilde{\mathbf{y}}_{i,-1} = (\tilde{y}_{i0}, \tilde{y}_{i1}, \dots, \tilde{y}_{i,T-1})'$, $\tilde{y}_{it} = y_{it} - N^{-1} \sum_{j=1}^N y_{jt}$, $t = 1, 2, \dots, T$. $E(\eta_T)$ and $\text{Var}(\eta_T)$ denote the mean and the variance of η_{iT} defined by (3.7). Suppose also that model (4.8)-(4.9) holds and the idiosyncratic shocks, ε_{it} , satisfy the Assumption 3.1 Then, under the null hypothesis, $\beta_i = 0$, $i = 1, 2, \dots, N$, as $N \rightarrow \infty$ and $T \rightarrow \infty$ (or $T \rightarrow \infty$, and $N \rightarrow \infty$)

$$\Gamma_{\widetilde{LM}} \Rightarrow N(0, 1).$$

Proof. See Section A.6 of the Appendix.

It is worth noting that our unit root test procedure for panels is likely to be more robust with respect to misspecification of time trends in the individual group regressions, than when the unit root test is applied to a single time series. This is because the common unobserved time-specific component can be removed in panels by the de-meaning procedure, but in the single time series case this is not possible. The panel unit-root test based on the de-meaned regressions, (4.10), will also be robust to general but common patterns of residual serial correlation across the groups.

The de-meaning procedure will not be, however, robust to misspecification of time trends or short-run dynamics if the effect of the common component varies across groups. As an example consider the following generalization of (4.9):

$$u_{it} = \lambda_i \theta_t + \varepsilon_{it}, \quad i = 1, \dots, N; \quad t = 1, \dots, T. \quad (4.13)$$

When θ_t is a known function of time (for example, $\theta_t = t$), this can be done by incorporating θ_t explicitly in the ADF regressions. But in the case where θ_t is not observed and λ_i differs across i , the analysis of unit roots in panel becomes much more complicated and requires further research.

5 Monte Carlo Simulation Results

In this section we use Monte Carlo experiments to examine finite sample properties of the alternative panel-based unit root tests. Here we also consider an alternative panel unit root test called the t -bar test, which is based on the average of the individual unit root t -statistics.

First, in the case where the disturbances in the underlying DF regressions are not serially correlated as in Section 3, the t -bar statistic is simply obtained by

$$\bar{t}_{NT} = \frac{1}{N} \sum_{i=1}^N t_{iT}, \quad (5.1)$$

where t_{iT} is the individual t statistic for testing $\beta_i = 0$ in (2.2). The exact sample critical values of the $\bar{t}_{NT}$ statistic can be computed via stochastic simulations, and are reported in Table 4. Also assuming that the second moments of t_{iT} , $i = 1, 2, \dots, N$, exist, under the null hypothesis (2.3) and Assumption 3.1, as $N \rightarrow \infty$, we have⁴

$$\Gamma_{\bar{t}} = \frac{\sqrt{N} \{\bar{t}_{NT} - E(t_T | \beta_i = 0)\}}{\sqrt{\text{Var}(t_T | \beta_i = 0)}} \Rightarrow N(0, 1), \quad (5.2)$$

where $E(t_T | \beta_i = 0)$ and $\text{Var}(t_T | \beta_i = 0)$ are the common mean and variance of t_{iT} , $i = 1, 2, \dots, N$, obtained under $\beta_i = 0$.

Next, in the case where the disturbances in the underlying DF regressions are serially correlated, we consider the above standardized t -bar statistic, and its modified version given by (which corresponds to (4.6))

$$\Psi_{\bar{t}} = \frac{\sqrt{N} \left\{ \bar{t}_{NT}(\mathbf{p}, \boldsymbol{\rho}) - \frac{1}{N} \sum_{i=1}^N E[t_{iT}(p_i, \mathbf{0}) | \beta_i = 0] \right\}}{\sqrt{\frac{1}{N} \sum_{i=1}^N \text{Var}[t_{iT}(p_i, \mathbf{0}) | \beta_i = 0]}}, \quad (5.3)$$

where $\bar{t}_{NT}(\mathbf{p}, \boldsymbol{\rho}) = \frac{1}{N} \sum_{i=1}^N t_{iT}(p_i, \boldsymbol{\rho}_i)$, $t_{iT}(p_i, \boldsymbol{\rho}_i)$ is the individual t statistic for testing $\beta_i = 0$ in (4.2). The values of $E[t_{iT}(p_i, \mathbf{0}) | \beta_i = 0]$ and $\text{Var}[t_{iT}(p_i, \mathbf{0}) | \beta_i = 0]$ have been evaluated via stochastic simulations, and are reported in Table 2. We expect $\Gamma_{\bar{t}}$ and $\Psi_{\bar{t}}$ to weakly converge to standard normal variates under the null hypothesis and diverge under the alternative hypothesis as both N and T tend to infinity such that $N/T \rightarrow k$, a finite positive constant.

We consider three sets of Monte Carlo experiments. The first set focuses on the benchmark model,

$$y_{it} = (1 - \phi_i) \mu_i + \phi_i y_{i,t-1} + \varepsilon_{it}, \quad t = 1, \dots, T; \quad i = 1, \dots, N. \quad (5.4)$$

The second set of experiments allows for the presence of positive (heterogeneous) AR(1) serial correlations in ε_{it} ,⁵

$$\varepsilon_{it} = \rho_i \varepsilon_{i,t-1} + e_{it}, \quad t = 1, \dots, T; \quad i = 1, \dots, N. \quad (5.5)$$

Finally, the third set of experiments allows for a linear trend in estimation of the ADF regressions. In all the three experiments e_{it} (or ε_{it} when $\rho_i = 0$) are generated as *iid* normal variates with zero means and heterogeneous variances, σ_i^2 , and μ_i , ρ_i and σ_i^2 are generated by

$$\mu_i \sim N(0, 1), \quad \rho_i \sim U[0.2, 0.4], \quad \sigma_i^2 \sim U[0.5, 1.5], \quad \text{for } i = 1, 2, \dots, N, \quad (5.6)$$

⁴We are not able to provide a formal derivation of the conditions under which the individual unit root t -statistics (t_{iT}) have second-order moments. However, on the basis of extensive stochastic simulations we conjecture that t_{iT} have finite second-order moments if $T \geq 5$, in the case of DF regressions containing an intercept only, and if $T \geq 6$, in the case of DF regressions containing an intercept as well as a linear time trend.

⁵We also experimented with positive and negative (heterogeneous) MA(1) as well as negative (heterogeneous) AR(1) serial correlations in ε_{it} . As far as the relative performance of the tests is concerned, we obtained similar results to those reported below.

where U stands for a uniform distribution, and μ_i , ρ_i and σ_i^2 are generated independently of ε_{it} . The first set of experiments were carried out for $N = 1, 5, 10, 25, 50, 100$, $T = 5, 10, 25, 50$. The other experiments were carried out for $N = 10, 25, 50, 100$, $T = 10, 25, 50$, and $p = 0, 1, 2, 3, 4$.⁶ We used 2,000 replications to compute empirical size and power of the tests at the 5% nominal level.

All experiments are carried out using the following statistics: the LM -bar, the t -bar and the LL tests. The results for the first set of experiments are summarized in Table 5.⁷ As a benchmark, in the first row of this table we give empirical size and power of the standard DF test. When $T = 5$ and 10, we report only the results for the LM -bar and the t -bar tests, because the adjustment factors needed to compute the LL statistic are not supplied by Levin and Lin (1993) for this case. The LM -bar and the t -bar tests ($\Gamma_{\overline{LM}}$ and $\Gamma_{\bar{t}}$ defined by (3.8) and (5.2), respectively) perform well even when $T = 5$. They have the correct size, and their power rises monotonically with N and T . For example, when $T = 10$, the power of the t -bar test increases steadily from 0.07 for $N = 5$ to 0.384 for $N = 100$. When $T = 5$, the power of the t -bar test increases with N but only very gradually. Overall, all the three tests have similar features, although for $T = 25$, the LL test seems to have a tendency to over-reject as N is allowed to increase. In addition, for small T the t -bar test has a slightly higher power even though the LL test has a larger size. The t -bar test also performs slightly better than the LM -bar test, especially for $N \geq 25$.

In the case of models with serially correlated errors, we experimented with both versions of the standardized LM -bar and the standardized t -bar tests; that is, $\Gamma_{\overline{LM}}$ and $\Gamma_{\bar{t}}$ defined in (3.8) and (5.2), and $\Psi_{\overline{LM}}$ and $\Psi_{\bar{t}}$ defined by (4.6) and (5.3), respectively. Although the tests are asymptotically equivalent, we have found that $\Psi_{\overline{LM}}$ and $\Psi_{\bar{t}}$ statistics, which explicitly take account of the underlying ADF orders in computing the mean and the variance adjustment factors, perform much better in small samples. Therefore, in what follows, we report only the results based on these statistics.

Table 6 summarizes the results for the second set of experiments. These results clearly show the importance of correctly choosing the order of the underlying ADF regressions. The problem is particularly serious when the order is underestimated. For example, incorrectly using the ADF(0) regressions to compute the test statistics causes the size of all the tests to go to zero.⁸ The situation is different, however, when the orders of the underlying ADF

⁶We also conducted simulation exercises where the ADF order is chosen by information criteria such as the Akaike Information or the Schwarz Criterion. The tests show a significant degree of size distortions, largely reflecting that in finite samples these criteria tend to select a low order for the ADF regressions. See also Ng and Perron (1995) for the effect of the ADF order selection on the finite sample performance of unit root tests in the case of single time series.

⁷We also carried out a number of experiments allowing for linear trends in the DF regressions and obtained similar results, although as to be expected the tests tended to have little power for T and N less than 25.

⁸The direction of the size distortion of the tests crucially depends on whether ε_{it} are positively or negatively autocorrelated. Failure to adequately allow for positive residual serial correlation results in under-rejection of the null hypothesis, while gross over-rejection results in the presence of neglected negative residual serial correlation. This is due to the fact that the distribution of the ADF(0) t -statistic gets shifted to the right with larger dispersions when there is a (neglected) positive residual serial correlation, and to the left with larger dispersions when there is a (neglected) negative residual serial correlation. Using 50,000 replications $E(t_T | \beta_i = 0)$ and $Var(t_T | \beta_i = 0)$ are simulated to be -1.5320 and 0.7346, respectively, when $\rho = 0$, while they are -0.8979 and 1.2789 when $\rho = 0.5$; and -2.6291 and 1.2061 when $\rho = -0.5$.

regressions are correctly chosen or over-estimated. In the case where the order is correctly chosen, the LM -bar and the t -bar tests, $\Psi_{\overline{LM}}$ and $\Psi_{\bar{t}}$, have empirical sizes very near to the nominal size, for all sample sizes, even when $T = 10$. In contrast, the LL test tends to over-reject the null hypothesis, and the extent of this over-rejection worsens as N increases. For example, when $T = 50$, the size of the LL test for $p = 1$ rises from 0.062 for $N = 10$ to 0.095 for $N = 100$. For $p > 1$, the size of the LL test tends towards zero as N increases.

Turning to the power performance of the tests, we find that the LM -bar and the t -bar tests are more powerful than the LL test, with the t -bar test having better small sample properties than the LM -bar test. For example, when $T = 50$ and $N = 25$, although the LL test for $p = 1$ is slightly over-sized (0.084) and the LM -bar and the t -bar test have correct sizes (0.057 and 0.064), the powers of the LM -bar and the t -bar test (0.912 and 0.972) substantially exceed that of the LL test (0.778).

In general, over-fitting is less harmful for inference on unit roots than under-fitting, but there is also a trade-off between power and size of the tests, which is a well-established empirical finding when testing for a unit root in the case of single time series. This trade-off is, however, much less favorable to the LL test than it is to the t -bar or the LM -bar tests. For example, when $T = 50$ and $N = 25$, the power of the t -bar test based on ADF(1) regressions decreases from 0.972 to 0.878 when ADF(4) regressions are used, but the power of the LL test drops from 0.778 to 0.026!

The results for the third set of experiments are summarized in Table 7. Once again an incorrect choice of the order of the ADF regression has detrimental effects for the performance of the test when the order is under-estimated. Correctly choosing the order of the ADF regression seems to be much more important in the present experiments where the panel contains deterministic time trends. In the case of the ADF(1) specification, the size of the t -bar test is around the nominal level even when $T = 10$. On the other hand, the LL test tends to under-reject, as N increases relative to T . This size distortion problem becomes more serious as p increases.

In general, both the LM -bar and the t -bar tests have better power performance than the LL test, and the t -bar test also tends to perform better than the LM -bar test. However, even the t -bar test has a rather poor power performance when $T = 10$. This is in line with the results obtained for the case of the ADF test applied to a single time series. Another important feature of the results lies in the fact that the power of the t -bar test is much more favorably affected by a rise in T than by an equivalent rise in N .⁹

⁹We also carried out a number of other experiments where the errors in (5.4) were generated with an unobservable common components. Once the common effect was eliminated by the de-meaning procedure discussed in subsection 4.1, the standardized LM -bar and the t -bar tests had their size close to the nominal level, and similar powers against homogeneous alternatives, $\beta_i = 0.9$ as obtained in the experiments without any common effects. However, when we considered heterogeneous alternatives where $\beta_i \sim U[0.85, 0.95]$, the de-meaning procedure adversely affected the power of the tests, as expected. A complete set of the Monte Carlo results for these and other experiments are available from the authors on request.

6 Concluding Remarks

In this paper we have developed a computationally simple procedure for testing the unit root hypothesis in heterogeneous panels. The small sample properties of the proposed tests are investigated via Monte Carlo methods. It is found that when there are no serial correlations, the LM -bar and the t -bar tests perform very well even when $T = 5$ or 10 . In this case it is possible to substantially augment the power of the unit root tests applied to single time series.

The situation is more complicated when the disturbances in the dynamic panel are serially correlated. The LM -bar and the t -bar test procedures now require that both N and T should be sufficiently large. Furthermore, in the case of serially correlated errors, it is critically important not to under-estimate the order of the underlying ADF regressions. When we allow for serial correlation and heterogeneity in the underlying data generating process, the simulation results clearly show that if a large enough order is selected for the underlying ADF regressions, then the finite sample performances of the LM -bar and the t -bar tests are reasonably satisfactory and much more superior to that of the LL test.

A Mathematical Appendix

A.1 Preliminary Results

We set out the following results, which will be used throughout this Appendix.

Lemma A.1 *Under Assumptions 4.1 - 4.2 and $\beta_i = 0$ in (2.2), as $T \rightarrow \infty$, the following results hold:*

$$T^{-1} \mathbf{W}_i' \mathbf{M}_\tau \mathbf{W}_i \xrightarrow{a.s.} \boldsymbol{\Omega}_i, \quad (\text{A.1})$$

$$T^{-1} \boldsymbol{\varepsilon}_i' \mathbf{M}_\tau \boldsymbol{\varepsilon}_i \xrightarrow{a.s.} \sigma_i^2 > 0, \quad (\text{A.2})$$

$$T^{-2} \mathbf{s}_{i,-1}' \mathbf{M}_\tau \mathbf{s}_{i,-1} \Rightarrow \sigma_i^2 \int_0^1 \overline{B}(r)^2 dr, \quad (\text{A.3})$$

$$T^{-1/2} \boldsymbol{\varepsilon}_i' \mathbf{M}_\tau \mathbf{W}_i \Rightarrow N(0, \sigma_i^2 \boldsymbol{\Omega}_i), \quad (\text{A.4})$$

$$T^{-1} \mathbf{s}_{i,-1}' \mathbf{M}_\tau \boldsymbol{\varepsilon}_i \Rightarrow \sigma_i^2 \int_0^1 \overline{B}(r) dB(r), \quad (\text{A.5})$$

$$T^{-1} \mathbf{s}_{i,-1}' \mathbf{M}_\tau \mathbf{W}_i \Rightarrow \sigma_i^2 \int_0^1 \overline{B}(r) d\overline{\mathbf{B}}_{\mathbf{W}_i}(r), \quad (\text{A.6})$$

where $\xrightarrow{a.s.}$ and $\Rightarrow$ signify the almost sure and weak convergence, respectively, $\mathbf{W}_i = (\Delta \mathbf{y}_{i,-1}, \Delta \mathbf{y}_{i,-2}, \dots, \Delta \mathbf{y}_{i,-p_i})$, $\boldsymbol{\Omega}_i$ is a $p_i \times p_i$ positive definite matrix, $B(r)$ is a scalar standard Brownian motion defined on $r \in (0, 1)$, $\mathbf{B}_{\mathbf{W}_i}(r)$ is a $p_i \times 1$ vector of Brownian motions with the positive definite covariance matrix $\boldsymbol{\Omega}_i$, $\overline{B}(r) = B(r) - \int_0^1 B(r) dr$ and $\overline{\mathbf{B}}_{\mathbf{W}_i}(r) = \mathbf{B}_{\mathbf{W}_i}(r) - \int_0^1 \mathbf{B}_{\mathbf{W}_i}(r) dr$. Note also that $\int_0^1 \overline{B}(r)^2 dr$ is strictly positive with probability 1.

Proof. See, for example, Hamilton (1994, Chapters 17 and 18).

Lemma A.2 *Let $\mathbf{z}_{iT}$, $i = 1, 2, \dots, N$, be a sequence of independent random variables such that $E(\mathbf{z}_{iT}^2) < \infty$, for all $T > T_0$. Then, as $N \rightarrow \infty$, $N^{-1} \sum_{i=1}^N \mathbf{z}_{iT} - N^{-1} \sum_{i=1}^N E(\mathbf{z}_{iT}) \xrightarrow{a.s.} 0$, and $\lim_{N \rightarrow \infty} N^{-1} \sum_{i=1}^N E(\mathbf{z}_{iT}) < \infty$.*

Proof. This is the special case of Corollary 3.9 in White (1984, p. 33).

Lemma A.3 *$\mathbf{D}$ is a $p \times p$ stochastic matrix; non-singular with probability one. Suppose that $\mathbf{D} = O_p(1)$ and $E(\mathbf{D}) = O(1)$. Then, $\mathbf{D}^{-1} = [E(\mathbf{D})]^{-1} + O_p(T^{-1/2})$.*

Proof. See Kiviet and Phillips (1994, Lemma 2).

Lemma A.4 *Let $\mathbf{v}$ be an $n \times 1$ vector of independent standard normal variables, and $\boldsymbol{\Gamma}_1$, $\boldsymbol{\Gamma}_2$ and $\boldsymbol{\Gamma}_3$ be non-stochastic (and possibly non-symmetric) $n \times n$ matrices then:*

$$\begin{aligned} E(\mathbf{v}' \boldsymbol{\Gamma}_1 \mathbf{v})(\mathbf{v}' \boldsymbol{\Gamma}_2 \mathbf{v}) &= tr(\boldsymbol{\Gamma}_1) tr(\boldsymbol{\Gamma}_2) + tr(\boldsymbol{\Gamma}_1 \boldsymbol{\Gamma}_2 + \boldsymbol{\Gamma}_1' \boldsymbol{\Gamma}_2), \\ E(\mathbf{v}' \boldsymbol{\Gamma}_1 \mathbf{v})(\mathbf{v}' \boldsymbol{\Gamma}_2 \mathbf{v})(\mathbf{v}' \boldsymbol{\Gamma}_3 \mathbf{v}) &= tr(\boldsymbol{\Gamma}_1) tr(\boldsymbol{\Gamma}_2) tr(\boldsymbol{\Gamma}_3) + 2tr(\boldsymbol{\Gamma}_1 \boldsymbol{\Gamma}_2 \boldsymbol{\Gamma}_3 + \boldsymbol{\Gamma}_1' \boldsymbol{\Gamma}_2 \boldsymbol{\Gamma}_3 + \boldsymbol{\Gamma}_1 \boldsymbol{\Gamma}_2' \boldsymbol{\Gamma}_3 + \boldsymbol{\Gamma}_1 \boldsymbol{\Gamma}_2 \boldsymbol{\Gamma}_3') \\ &\quad + tr(\boldsymbol{\Gamma}_1) tr(\boldsymbol{\Gamma}_2 \boldsymbol{\Gamma}_3 + \boldsymbol{\Gamma}_2' \boldsymbol{\Gamma}_3) + tr(\boldsymbol{\Gamma}_2) tr(\boldsymbol{\Gamma}_1 \boldsymbol{\Gamma}_3 + \boldsymbol{\Gamma}_1' \boldsymbol{\Gamma}_3) + tr(\boldsymbol{\Gamma}_3) tr(\boldsymbol{\Gamma}_1 \boldsymbol{\Gamma}_2 + \boldsymbol{\Gamma}_1' \boldsymbol{\Gamma}_2), \end{aligned}$$

where 'tr' denotes the trace of the matrix.

Proof. See Lemma 1 in Kiviet and Phillips (1996, Appendix A). Though Kiviet and Phillips do not consider the case where $\boldsymbol{\Gamma}_1$, $\boldsymbol{\Gamma}_2$ and $\boldsymbol{\Gamma}_3$ are all non-symmetric, the above result can be easily obtained noting that $\mathbf{v}' \mathbf{A} \mathbf{v} = \mathbf{v}' \frac{\mathbf{A} + \mathbf{A}'}{2} \mathbf{v}$ for any non-symmetric matrix $\mathbf{A}$. The detailed proof will be available upon request.

Lemma A.5 Let $\mathbf{v}$ be an $n \times 1$ vector of independent normal variables, and Λ be an $n \times n$ positive semi-definite symmetric matrix of rank g . Then the r -th moment of the inverse of $\mathbf{v}'\Lambda\mathbf{v}$ exists if $g > 2r$.

Proof. See Smith (1988).

Corollary A.1 Let Λ_1 and Λ_2 be $n \times n$ positive semi-definite symmetric matrices of rank g_1 and g_2 , respectively. Then the r -th moment of the inverse of $(\mathbf{v}'\Lambda_1\mathbf{v})(\mathbf{v}'\Lambda_2\mathbf{v})$ exists if $g_1 > 4r$ and $g_2 > 4r$.

Proof. The result is immediate from Lemma 1 using the Cauchy-Schwarz inequality.

Lemma A.6 The r -th moment of U/V exists if $E(U^r)$ and $E(1/V^{r+\epsilon})$ exist for any $\epsilon > 0$.

Proof. The result follows from the Hölder's inequality.

Lemma A.7 Suppose X_{iNT} have finite first- and second-order moments for all i , N and T , and that $E(X_{iNT}) = o(1)$ and $Var(X_{iNT}) = o(1)$ for all i . Then,

$$\bar{X}_{NT} = N^{-1} \sum_{i=1}^N X_{iNT} = o_p(1).$$

Proof. Since X_{iNT} have finite first-order moments then

$$E|\bar{X}_{NT}| \leq N^{-1} \sum_{i=1}^N E|X_{iNT}| \leq \max_{\ell} E|X_{\ell NT}| = o(1).$$

Also,

$$Var(\bar{X}_{NT}) = N^{-2} \left(\sum_{i=1}^N Var(X_{iNT}) + \sum_{j \neq i} \sum_{i=1}^N Cov(X_{iNT}, X_{jNT}) \right) \leq \max_{\ell} Var(X_{\ell NT}) = o(1),$$

since $Cov(X_{iNT}, X_{jNT}) \leq \sqrt{Var(X_{iNT})Var(X_{jNT})} \leq \max_{\ell} Var(X_{\ell NT})$. Therefore, $\bar{X}_{NT}$ converges to zero in mean squared errors which in turn implies $\bar{X}_{NT} = o_p(1)$.

A.2 Proof of Theorem 3.1

Under $\beta_i = 0$, $LM_{iT} = \eta_{iT}$. Since η_{iT} , $i = 1, 2, \dots, N$, are identically and independently distributed with a finite mean $E(\eta_T)$ and a finite variance $Var(\eta_T)$, hence by the Lindberg-Levy central limit theorem (CLT), under the null (2.3) and Assumption 3.1, as $N \rightarrow \infty$,

$$\sqrt{N}(\bar{LM}_{NT} - E(\eta_T)) = \frac{1}{\sqrt{N}} \sum_{i=1}^N (\eta_{iT} - E(\eta_T)) \Rightarrow N\{0, Var(\eta_T)\}, \quad (\text{A.7})$$

and therefore,

$$\Gamma_{\bar{LM}} = \frac{\sum_{i=1}^N \{\eta_{iT} - E(\eta_T)\}}{\sqrt{NVar(\eta_T)}} \Rightarrow N(0, 1). \quad (\text{A.8})$$

A.3 On the consistency of the LM -bar test - the case of the serially uncorrelated errors

To establish the conditions under which the test based on $\Gamma_{\overline{LM}}$ is likely to be consistent, we first notice that $\mathbf{y}_{i,-1}$, $\mathbf{P}_i$, $\boldsymbol{\xi}_i$ and other notations used later all depend on β_i when $\beta_i < 0$. But to simplify the analysis we do not make their dependence explicit except when this is needed.

Using (3.2) and after some algebra we have

$$LM_{iT} = \frac{T(\boldsymbol{\varepsilon}'_i \mathbf{P}_i \boldsymbol{\varepsilon}_i + \beta_i A_{iT})}{\boldsymbol{\varepsilon}'_i \mathbf{M}_\tau \boldsymbol{\varepsilon}_i + \beta_i A_{iT}}, \quad (\text{A.9})$$

where $A_{iT} = \beta_i (\mathbf{y}'_{i,-1} \mathbf{M}_\tau \mathbf{y}_{i,-1}) + 2(\mathbf{y}'_{i,-1} \mathbf{M}_\tau \boldsymbol{\varepsilon}_i)$. Letting

$$h_{iT}(\beta_i) = \frac{T\boldsymbol{\varepsilon}'_i \mathbf{P}_i \boldsymbol{\varepsilon}_i}{\boldsymbol{\varepsilon}'_i \mathbf{M}_\tau \boldsymbol{\varepsilon}_i} = \frac{T(\boldsymbol{\varepsilon}'_i \mathbf{M}_\tau \mathbf{y}_{i,-1})^2}{(\boldsymbol{\varepsilon}'_i \mathbf{M}_\tau \boldsymbol{\varepsilon}_i)(\mathbf{y}'_{i,-1} \mathbf{M}_\tau \mathbf{y}_{i,-1})}, \quad (\text{A.10})$$

then

$$LM_{iT} = h_{iT}(\beta_i) + T\beta_i k_{iT}, \quad (\text{A.11})$$

where

$$k_{iT} = A_{iT} \frac{(\boldsymbol{\varepsilon}'_i \mathbf{M}_\tau \boldsymbol{\varepsilon}_i) - (\boldsymbol{\varepsilon}'_i \mathbf{P}_i \boldsymbol{\varepsilon}_i)}{(\boldsymbol{\varepsilon}'_i \mathbf{M}_\tau \boldsymbol{\varepsilon}_i)(\boldsymbol{\xi}'_i \mathbf{M}_\tau \boldsymbol{\xi}_i)}.$$

Solving (2.1) iteratively, we have

$$y_{it} = \phi_i^t y_{i0} + (1 - \phi_i^t) \mu_i + \sum_{j=0}^{t-1} \phi_i^j \varepsilon_{i,t-j} = y_{i0} + s_{it} + (1 - \phi_i^t)(\mu_i - y_{i0}) - \sum_{j=0}^{t-1} (1 - \phi_i^j) \varepsilon_{i,t-j}, \quad (\text{A.12})$$

where $s_{it} = \sum_{j=1}^t \varepsilon_{ij}$. Similarly,

$$y_{i,t-1} = y_{i0} + s_{i,t-1} + (1 - \phi_i^{t-1})(\mu_i - y_{i0}) - \sum_{j=0}^{t-1} (1 - \phi_i^j) \varepsilon_{i,t-1-j}. \quad (\text{A.13})$$

Using $(1 - \phi_i) = -\beta_i$ and $(1 - \phi_i^{t-1}) = (1 - \phi_i)(1 + \phi_i + \dots + \phi_i^{t-2})$ in (A.13), then

$$y_{i,t-1} = y_{i0} + s_{i,t-1} + \beta_i q_{i,t-1}, \quad (\text{A.14})$$

where

$$q_{i,t-1} = -(1 + \phi_i + \dots + \phi_i^{t-2})(\mu_i - y_{i0}) + \sum_{j=0}^{t-1} (1 + \phi_i + \dots + \phi_i^{j-1}) \varepsilon_{i,t-1-j}, \quad (\text{A.15})$$

or in the matrix notation,

$$\mathbf{y}_{i,-1} = y_{i0} \boldsymbol{\tau}_T + \mathbf{s}_{i,-1} + \beta_i \mathbf{q}_{i,-1}, \quad (\text{A.16})$$

where $\mathbf{q}_{i,-1} = (q_{i0}, q_{i1}, \dots, q_{i,T-1})'$.

Using this result in (A.10) now yields

$$h_{iT}(\beta_i) = \frac{T(\boldsymbol{\varepsilon}'_i \mathbf{M}_\tau \mathbf{s}_{i,-1})^2 + T\beta_i W_{iT}}{(\boldsymbol{\varepsilon}'_i \mathbf{M}_\tau \boldsymbol{\varepsilon}_i) \{(\mathbf{s}_{i,-1} \mathbf{M}_\tau \mathbf{s}_{i,-1}) + \beta_i V_{iT}\}}, \quad (\text{A.17})$$

where $W_{iT} = \beta_i (\boldsymbol{\varepsilon}'_i \mathbf{M}_\tau \mathbf{q}_{i,-1})^2 + 2(\boldsymbol{\varepsilon}'_i \mathbf{M}_\tau \mathbf{s}_{i,-1})(\boldsymbol{\varepsilon}'_i \mathbf{M}_\tau \mathbf{q}_{i,-1})$ and $V_{iT} = \beta_i (\mathbf{q}'_{i,-1} \mathbf{M}_\tau \mathbf{q}_{i,-1}) + 2(\mathbf{q}'_{i,-1} \mathbf{M}_\tau \mathbf{s}_{i,-1})$. Therefore,

$$h_{iT}(\beta_i) = \eta_{iT} + T\beta_i f_{iT}, \quad (\text{A.18})$$

where η_{iT} is defined by (3.5),

$$f_{iT} = \frac{W_{iT} (\mathbf{s}'_{i,-1} \mathbf{M}_\tau \mathbf{s}_{i,-1}) - V_{iT} (\boldsymbol{\varepsilon}'_i \mathbf{M}_\tau \mathbf{s}_{i,-1})^2}{(\boldsymbol{\varepsilon}'_i \mathbf{M}_\tau \boldsymbol{\varepsilon}_i) (\mathbf{s}'_{i,-1} \mathbf{M}_\tau \mathbf{s}_{i,-1}) (\mathbf{y}'_{i,-1} \mathbf{M}_\tau \mathbf{y}_{i,-1})} = \frac{W_{iT} - V_{iT} (\boldsymbol{\varepsilon}'_i \mathbf{P}_i(0) \boldsymbol{\varepsilon}_i)}{(\boldsymbol{\varepsilon}'_i \mathbf{M}_\tau \boldsymbol{\varepsilon}_i) (\mathbf{y}'_{i,-1} \mathbf{M}_\tau \mathbf{y}_{i,-1})}, \quad (\text{A.19})$$

and $\mathbf{P}_i(0) = \mathbf{M}_\tau \mathbf{s}_{i,-1} (\mathbf{s}'_{i,-1} \mathbf{M}_\tau \mathbf{s}_{i,-1})^{-1} \mathbf{s}'_{i,-1} \mathbf{M}_\tau$. Using (A.18) in (A.11) we have

$$LM_{iT} - \eta_{iT} = T\beta_i g_{iT}, \quad (\text{A.20})$$

where $g_{iT} = f_{iT} + k_{iT}$. It is now easily seen that the power of the test will, in general, depend on $(y_{i0} - \mu_i) / \sigma_i$ when T is finite.¹⁰ Under the alternative hypothesis (2.4) and using (A.20), the standardized LM -bar statistic, defined by (3.8), can be written as

$$\Gamma_{LM} = \frac{\sum_{i=1}^N \{\eta_{iT} - E(\eta_T)\}}{\sqrt{N \text{Var}(\eta_T)}} + \left(\frac{T\sqrt{N}}{\text{Var}(\eta_T)} \right) \left(\frac{N_1 \sum_{i=1}^{N_1} \beta_i g_{iT}}{N_1} \right). \quad (\text{A.21})$$

Since LM_{iT} and η_{iT} have second moments, so will g_{iT} for $\beta_i < 0$. Further, noticing that g_{iT} are independently (but not identically) distributed across i , then by the strong law of large numbers (see Lemma A.2) $N_1^{-1} \sum_{i=1}^{N_1} \beta_i g_{iT} - N_1^{-1} \sum_{i=1}^{N_1} \beta_i E(g_{iT}) \xrightarrow{a.s.} 0$. In view of this result, using (A.8) and under the alternative hypothesis (2.4), as N and $N_1 \rightarrow \infty$,

$$\text{plim}_{N \rightarrow \infty} \left(N^{-1/2} \Gamma_{LM} \right) = \frac{T\delta_1}{\sqrt{\text{Var}(\eta_T)}} \lim_{N_1 \rightarrow \infty} \left\{ \frac{\sum_{i=1}^{N_1} \beta_i E(g_{iT})}{N_1} \right\},$$

where $\delta_1 = \lim_{N \rightarrow \infty} (N_1/N) > 0$.

Using (A.20) we now have

$$\text{plim}_{N \rightarrow \infty} \left(N^{-1/2} \Gamma_{LM} \right) = \frac{T\delta_1}{\sqrt{\text{Var}(\eta_T)}} \lim_{N_1 \rightarrow \infty} \left\{ N_1^{-1} \sum_{i=1}^{N_1} [E(LM_{iT} | \beta_i < 0) - E(\eta_T)] \right\}.$$

Since LM_{iT} and η_{iT} both have finite second-order moments, values of $E(LM_{iT} | \beta_i < 0)$ and $E(\eta_T)$ can be obtained via stochastic simulations. Under our assumptions $E(\eta_T)$ for different values of T are given in Table 1. The values of $E(LM_{iT} | \beta_i < 0)$ in general depend on T , β_i and $(y_{i0} - \mu_i) / \sigma_i$. Our computations suggest that so long as y_{i0} does not deviate too much from its mean μ_i , say $y_{i0} = \mu_i \pm 3\sigma_i$, then $E(LM_{iT} | \beta_i < 0) > E(\eta_T)$.¹¹ But for extremely large values of y_{i0} , say $|y_{i0} - \mu_i| > 3\sigma_i$, it is possible for $E(\eta_T)$ to exceed $E(LM_{iT} | \beta_i < 0)$.¹²

A.4 Proof of Theorem 4.1

A.4.1 Asymptotic distribution under the null

Under $\beta_i = 0$, Δy_{it} 's are generated by stationary $AR(p_i)$ processes,

$$\Delta y_{it} = \sum_{j=1}^{p_i} \rho_{ij} \Delta y_{i,t-j} + \varepsilon_{it}, \quad i = 1, 2, \dots, N; \quad t = 1, 2, \dots, T, \quad (\text{A.22})$$

which can be expressed in the matrix form as

$$\Delta y_i = \mathbf{W}_i \boldsymbol{\rho}_i + \boldsymbol{\varepsilon}_i, \quad (\text{A.23})$$

¹⁰DeJong *et al.* (1992) have also shown that for a finite T the unit root test applied to the single time series depends on $(y_{i0} - \mu_i) / \sigma_i$ under the alternative of (trend) stationarity.

¹¹Under the assumption that $y_{i0} \sim N(\mu_i, \sigma_i^2)$, the probability that $|y_{i0} - \mu_i| > 3\sigma_i$, is less than 0.0026.

¹²The values of $E(LM_{iT} | \beta_i < 0) - E(\eta_T)$ tabulated for different values of T , β_i and $(y_{i0} - \mu_i) / \sigma_i$ can be obtained from the authors on request.

where $\mathbf{W}_i = (\Delta \mathbf{y}_{i,-1}, \Delta \mathbf{y}_{i,-2}, \dots, \Delta \mathbf{y}_{i,-p_i})$, $\boldsymbol{\rho}_i = (\rho_{i1}, \rho_{i2}, \dots, \rho_{ip_i})'$ and $\boldsymbol{\varepsilon}_i = (\varepsilon_{i1}, \varepsilon_{i2}, \dots, \varepsilon_{iT})'$. Next, summing (A.22) from 1 to t we have

$$s_{it} = y_{it} - \sum_{j=1}^{p_i} \rho_{ij} \Delta y_{i,t-j} - \left(y_{i0} - \sum_{j=1}^{p_i} \rho_{ij} y_{i,-j} \right),$$

where $s_{it} = \sum_{j=1}^t \varepsilon_{ij}$. Hence,

$$s_{i,t-1} = \left(1 - \sum_{j=1}^{p_i} \rho_{ij} \right) y_{i,t-1} + \sum_{j=1}^{p_i} \varphi_{ij} \Delta y_{i,t-j} - \left(y_{i0} - \sum_{j=1}^{p_i} \rho_{ij} y_{i,-j} \right),$$

where $\varphi_{ij} = \sum_{h=j}^{p_i} \rho_{ih}$. Since $1 - \sum_{j=1}^{p_i} \rho_{ij} \neq 0$ by Assumption 4.1, then

$$\mathbf{y}_{i,-1} = \left(\frac{1}{1 - \sum_{j=1}^{p_i} \rho_{ij}} \right) (\mathbf{s}_{i,-1} - \mathbf{Q}_i \boldsymbol{\varphi}_i), \quad (\text{A.24})$$

where $\boldsymbol{\varphi}_i = (\varphi_{i0}, \varphi_{i1}, \dots, \varphi_{ip_i})'$, and $\varphi_{i0} = -y_{i0} + \sum_{j=1}^{p_i} \rho_{ij} y_{i,-j}$, which we assume as given (see Assumption 4.2). Using (A.23) and (A.24), and noting that $\mathbf{W}_i \subset \mathbf{Q}_i = (\boldsymbol{\tau}_T, \mathbf{W}_i)$, then under $\beta_i = 0$,

$$\Delta \mathbf{y}_i' \mathbf{P}_i^* \Delta \mathbf{y}_i = (\boldsymbol{\varepsilon}_i' \mathbf{M}_{Q_i} \mathbf{s}_{i,-1}) (\mathbf{s}_{i,-1}' \mathbf{M}_{Q_i} \mathbf{s}_{i,-1})^{-1} (\mathbf{s}_{i,-1}' \mathbf{M}_{Q_i} \boldsymbol{\varepsilon}_i), \quad \Delta \mathbf{y}_i' \mathbf{M}_{Q_i} \Delta \mathbf{y}_i = \boldsymbol{\varepsilon}_i' \mathbf{M}_{Q_i} \boldsymbol{\varepsilon}_i,$$

and therefore, $LM_{iT}(p_i, \boldsymbol{\rho}_i)$, which for expositional simplicity we denote by d_{iT} , can be written as

$$d_{iT} = \frac{T (\boldsymbol{\varepsilon}_i' \mathbf{M}_{Q_i} \mathbf{s}_{i,-1})^2}{(\boldsymbol{\varepsilon}_i' \mathbf{M}_{Q_i} \boldsymbol{\varepsilon}_i) (\mathbf{s}_{i,-1}' \mathbf{M}_{Q_i} \mathbf{s}_{i,-1})}. \quad (\text{A.25})$$

Our objective here is to derive the asymptotic distribution of $\Gamma_{\overline{LM}}$ defined by (3.8) under the null hypothesis, or equivalently the asymptotic distribution of

$$\sqrt{N} \{ \bar{d}_{NT} - E(\eta_T) \} = \sqrt{N} (\bar{d}_{NT} - \bar{\eta}_{NT}) + \sqrt{N} (\bar{\eta}_{NT} - E(\eta_T)) \quad (\text{A.26})$$

Initially, we focus on the first term in the above decomposition. But first we need a number of preliminary results.

Since $\mathbf{M}_{Q_i} = \mathbf{M}_\tau - \tilde{\mathbf{P}}_i$, where $\tilde{\mathbf{P}}_i = \mathbf{M}_\tau \mathbf{W}_i (\mathbf{W}_i' \mathbf{M}_\tau \mathbf{W}_i)^{-1} \mathbf{W}_i' \mathbf{M}_\tau$, then it is easily seen that

$$\boldsymbol{\varepsilon}_i' \mathbf{M}_{Q_i} \boldsymbol{\varepsilon}_i = \boldsymbol{\varepsilon}_i' \mathbf{M}_\tau \boldsymbol{\varepsilon}_i (1 - T^{-1} \lambda_{iT}), \quad \mathbf{s}_{i,-1}' \mathbf{M}_{Q_i} \mathbf{s}_{i,-1} = \mathbf{s}_{i,-1}' \mathbf{M}_\tau \mathbf{s}_{i,-1} (1 - T^{-1} \theta_{iT}),$$

where

$$\lambda_{iT} = \frac{(T^{-1/2} \boldsymbol{\varepsilon}_i' \mathbf{M}_\tau \mathbf{W}_i) (T^{-1} \mathbf{W}_i' \mathbf{M}_\tau \mathbf{W}_i)^{-1} (T^{-1/2} \mathbf{W}_i' \mathbf{M}_\tau \boldsymbol{\varepsilon}_i)}{T^{-1} \boldsymbol{\varepsilon}_i' \mathbf{M}_\tau \boldsymbol{\varepsilon}_i}, \quad (\text{A.27})$$

$$\theta_{iT} = \frac{(T^{-1} \mathbf{s}_{i,-1}' \mathbf{M}_\tau \mathbf{W}_i) (T^{-1} \mathbf{W}_i' \mathbf{M}_\tau \mathbf{W}_i)^{-1} (T^{-1} \mathbf{W}_i' \mathbf{M}_\tau \mathbf{s}_{i,-1})}{T^{-2} \mathbf{s}_{i,-1}' \mathbf{M}_\tau \mathbf{s}_{i,-1}}. \quad (\text{A.28})$$

Therefore, we have

$$\begin{aligned} d_{iT} &= \frac{T \left(\boldsymbol{\varepsilon}_i' \mathbf{M}_\tau \mathbf{s}_{i,-1} - \boldsymbol{\varepsilon}_i' \tilde{\mathbf{P}}_i \mathbf{s}_{i,-1} \right)^2}{(\boldsymbol{\varepsilon}_i' \mathbf{M}_\tau \boldsymbol{\varepsilon}_i) (\mathbf{s}_{i,-1}' \mathbf{M}_\tau \mathbf{s}_{i,-1})} \times \frac{1}{(1 - T^{-1} \lambda_{iT})(1 - T^{-1} \theta_{iT})} \\ &= \eta_{iT} - \frac{2}{\sqrt{T}} H_{iT} + \frac{1}{T} (G_{iT} + F_{iT}), \end{aligned} \quad (\text{A.29})$$

where

$$H_{iT} = \frac{(T^{-1} \boldsymbol{\varepsilon}_i' \mathbf{M}_\tau \mathbf{s}_{i,-1}) (T^{-1/2} \boldsymbol{\varepsilon}_i' \mathbf{M}_\tau \mathbf{W}_i) (T^{-1} \mathbf{W}_i' \mathbf{M}_\tau \mathbf{W}_i)^{-1} (T^{-1} \mathbf{W}_i' \mathbf{M}_\tau \mathbf{s}_{i,-1})}{(T^{-1} \boldsymbol{\varepsilon}_i' \mathbf{M}_\tau \boldsymbol{\varepsilon}_i) (T^{-2} \mathbf{s}_{i,-1}' \mathbf{M}_\tau \mathbf{s}_{i,-1})}, \quad (\text{A.30})$$

[A.5]

$$G_{iT} = \frac{\left[(T^{-1/2} \boldsymbol{\varepsilon}'_i \mathbf{M}_\tau \mathbf{W}_i) (T^{-1} \mathbf{W}'_i \mathbf{M}_\tau \mathbf{W}_i)^{-1} (T^{-1} \mathbf{W}'_i \mathbf{M}_\tau \mathbf{s}_{i,-1}) \right]^2}{(T^{-1} \boldsymbol{\varepsilon}'_i \mathbf{M}_\tau \boldsymbol{\varepsilon}_i) (T^{-2} \mathbf{s}'_{i,-1} \mathbf{M}_\tau \mathbf{s}_{i,-1})}, \quad (\text{A.31})$$

$$F_{iT} = \left(\eta_{iT} - \frac{2}{\sqrt{T}} H_{iT} + \frac{1}{T} G_{iT} \right) \left(\frac{\lambda_{iT}}{1 - T^{-1} \lambda_{iT}} + \frac{\theta_{iT}}{1 - T^{-1} \theta_{iT}} + \frac{T^{-1} \lambda_{iT} \theta_{iT}}{(1 - T^{-1} \lambda_{iT})(1 - T^{-1} \theta_{iT})} \right). \quad (\text{A.32})$$

Using (A.25) it is now easily seen that

$$d_{iT} = \frac{(T^{-1} \boldsymbol{\varepsilon}'_i \mathbf{M}_{Q_i} \mathbf{s}_{i,-1}) (T^{-2} \mathbf{s}'_{i,-1} \mathbf{M}_{Q_i} \mathbf{s}_{i,-1})^{-1} (T^{-1} \mathbf{s}'_{i,-1} \mathbf{M}_{Q_i} \boldsymbol{\varepsilon}_i)}{(T^{-1} \boldsymbol{\varepsilon}'_i \mathbf{M}_{Q_i} \boldsymbol{\varepsilon}_i)},$$

and¹³

$$d_{iT} \leq T \frac{\|\boldsymbol{\varepsilon}'_i \mathbf{M}_{Q_i}\|^2 \left\| \mathbf{M}_{Q_i} \mathbf{s}_{i,-1} (\mathbf{s}'_{i,-1} \mathbf{M}_{Q_i} \mathbf{s}_{i,-1})^{-1} (\mathbf{s}'_{i,-1} \mathbf{M}_{Q_i}) \right\|}{\|\boldsymbol{\varepsilon}'_i \mathbf{M}_{Q_i} \boldsymbol{\varepsilon}_i\|} = T.$$

Similarly, $\lambda_{iT} \leq T$, $\theta_{iT} \leq T$, $H_{iT} \leq T$ and $G_{iT} \leq T$. Therefore, all the moments of d_{iT} , λ_{iT} , θ_{iT} , H_{iT} and G_{iT} exist when T is finite. Also, since $F_{iT} = T d_{iT} - T \eta_{iT} + 2\sqrt{T} H_{iT} - T G_{iT}$ and η_{iT} has moments, hence all the moments of F_{iT} exist when T is finite. Furthermore, using the standard results summarized in Lemma A.1 it is also easily seen that $d_{iT} = O_p(1)$, $\eta_{iT} = O_p(1)$, $\lambda_{iT} = O_p(1)$, $\theta_{iT} = O_p(1)$, $H_{iT} = O_p(1)$, $G_{iT} = O_p(1)$ and $F_{iT} = O_p(1)$; and therefore, d_{iT} , λ_{iT} , θ_{iT} , H_{iT} , G_{iT} and F_{iT} have moments even as $T \rightarrow \infty$, in particular, $E(G_{iT}) = O(1)$ and $E(F_{iT}) = O(1)$.

In view of the above results on existence of moments and averaging (A.29) across $i = 1, 2, \dots, N$, the first term in (A.26) becomes

$$\begin{aligned} \sqrt{N} (\bar{d}_{NT} - \bar{\eta}_{NT}) &= -2\sqrt{\frac{N}{T}} \left[\frac{1}{N} \sum_{i=1}^N E(H_{iT}) \right] + \frac{\sqrt{N}}{T} \left[\frac{1}{N} \sum_{i=1}^N E(G_{iT} + F_{iT}) \right] \\ &\quad - 2\sqrt{\frac{N}{T}} \left\{ \frac{1}{N} \sum_{i=1}^N [H_{iT} - E(H_{iT})] \right\} + \frac{\sqrt{N}}{T} \left\{ \frac{1}{N} \sum_{i=1}^N [G_{iT} + F_{iT} - E(G_{iT} + F_{iT})] \right\} \end{aligned} \quad (\text{A.33})$$

Now by strong law of large number in Lemma A.2, we have

$$\frac{1}{N} \sum_{i=1}^N [H_{iT} - E(H_{iT})] \xrightarrow{a.s.} 0, \quad \frac{1}{N} \sum_{i=1}^N [G_{iT} + F_{iT} - E(G_{iT} + F_{iT})] \xrightarrow{a.s.} 0. \quad (\text{A.34})$$

Hence, (noting that by assumption the limit of N/T is finite as N and $T \rightarrow \infty$)

$$\sqrt{N} (\bar{d}_{NT} - \bar{\eta}_{NT}) = -2\sqrt{\frac{N}{T}} \left[\frac{1}{N} \sum_{i=1}^N E(H_{iT}) \right] + \frac{\sqrt{N}}{T} \left[\frac{1}{N} \sum_{i=1}^N E(G_{iT} + F_{iT}) \right] + o_p(1). \quad (\text{A.35})$$

Since moments of G_{iT} and F_{iT} exist for any T , and under our assumptions $\sqrt{N}/T \rightarrow 0$, then the second term in the above expression tends to zero as N and $T \rightarrow \infty$.

Consider now the limit of $N^{-1} \sum_{i=1}^N E(H_{iT})$ as both $T \rightarrow \infty$ and $N \rightarrow \infty$. Using Lemma A.3 we have $(T^{-1} \boldsymbol{\varepsilon}'_i \mathbf{M}_\tau \boldsymbol{\varepsilon}_i)^{-1} = [E(T^{-1} \boldsymbol{\varepsilon}'_i \mathbf{M}_\tau \boldsymbol{\varepsilon}_i)]^{-1} + O_p(T^{-1/2})$, $(T^{-2} \mathbf{s}'_{i,-1} \mathbf{M}_\tau \mathbf{s}_{i,-1})^{-1} = [E(T^{-2} \mathbf{s}'_{i,-1} \mathbf{M}_\tau \mathbf{s}_{i,-1})]^{-1} + O_p(T^{-1/2})$, and $(T^{-1} \mathbf{W}'_i \mathbf{M}_\tau \mathbf{W}_i)^{-1} = [E(T^{-1} \mathbf{W}'_i \mathbf{M}_\tau \mathbf{W}_i)]^{-1} + O_p(T^{-1/2})$, where $E(T^{-1} \boldsymbol{\varepsilon}'_i \mathbf{M}_\tau \boldsymbol{\varepsilon}_i) = O(1)$, $E(T^{-2} \mathbf{s}'_{i,-1} \mathbf{M}_\tau \mathbf{s}_{i,-1}) = O(1)$ and $E(T^{-1} \mathbf{W}'_i \mathbf{M}_\tau \mathbf{W}_i) = O(1)$. Therefore,

$$H_{iT} = K_{iT} + \frac{R_{iT}}{\sqrt{T}},$$

¹³Note that here the norm of matrix $\mathbf{A}$ is defined as $\|\mathbf{A}\| = [\text{tr}(\mathbf{A}\mathbf{A}')]^{1/2}$.

where $R_{iT} = O_p(1)$ and has all moments when T is finite, and

$$K_{iT} = \frac{(T^{-1}\boldsymbol{\varepsilon}'_i \mathbf{M}_\tau \mathbf{s}_{i,-1}) (T^{-1/2}\boldsymbol{\varepsilon}'_i \mathbf{M}_\tau \mathbf{W}_i) [E(T^{-1}\mathbf{W}'_i \mathbf{M}_\tau \mathbf{W}_i)]^{-1} (T^{-1}\mathbf{W}'_i \mathbf{M}_\tau \mathbf{s}_{i,-1})}{E(T^{-1}\boldsymbol{\varepsilon}'_i \mathbf{M}_\tau \boldsymbol{\varepsilon}_i) E(T^{-2}\mathbf{s}'_{i,-1} \mathbf{M}_\tau \mathbf{s}_{i,-1})}.$$

Again by Lemma A.2, $N^{-1} \sum_{i=1}^N R_{iT} - N^{-1} \sum_{i=1}^N E(R_{iT}) \xrightarrow{a.s.} 0$, and therefore, $T^{-1/2} \left(\frac{1}{N} \sum_{i=1}^N R_{iT} \right) \xrightarrow{a.s.} 0$. Hence, $N^{-1} \sum_{i=1}^N H_{iT} - N^{-1} \sum_{i=1}^N K_{iT} \xrightarrow{a.s.} 0$. But, as it is shown in Section A.5,

$$E(K_{iT}) = O(T^{-1/2}). \quad (\text{A.36})$$

Hence, $\lim_{T \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N E(H_{iT}) = \lim_{T \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N E(K_{iT}) = 0$. Using this result in (A.35), and noting that by assumption N/T tends to a finite limit, we have

$$\sqrt{N} (\bar{d}_{NT} - \bar{\eta}_{NT}) = o_p(1). \quad (\text{A.37})$$

It now remains to derive the asymptotic distribution of $\sqrt{N} \{\bar{\eta}_{NT} - E(\eta_T)\}$. By the central limit theorem for triangular arrays [see Billingsley (1979, Theorem 27.2)], as $T \rightarrow \infty$, and $N \rightarrow \infty$, we have

$$\frac{\sqrt{N} \{\bar{\eta}_{NT} - E(\eta_T)\}}{\sqrt{\text{Var}(\eta_T)}} \Rightarrow N(0, 1). \quad (\text{A.38})$$

Combining (A.37) and (A.38), under the null hypothesis and Assumptions 4.1 - 4.2, as T and $N \rightarrow \infty$, such that $N/T \rightarrow k$, where k is a finite (positive) constant, we have

$$\Gamma_{LM} = \frac{\sqrt{N} \{\overline{LM}_{NT} - E(\eta_T)\}}{\sqrt{\text{Var}(\eta_T)}} \Rightarrow N(0, 1). \quad (\text{A.39})$$

A.4.2 Proof of the consistency

Under $\beta_i < 0$ we have

$$\Delta \mathbf{y}_i = \beta_i \mathbf{y}_{i,-1} + \mathbf{Q}_i \boldsymbol{\gamma}_i + \boldsymbol{\varepsilon}_i = \boldsymbol{\xi}_i + \boldsymbol{\varepsilon}_i. \quad (\text{A.40})$$

The expressions for $\Delta \mathbf{y}_i$, $\mathbf{Q}_i$ and $\boldsymbol{\xi}_i$ now depend on β_i . Since $\mathbf{M}_{Q_i} \mathbf{Q}_i = \mathbf{0}$ and $\mathbf{P}_i^* \mathbf{Q}_i = \mathbf{0}$, hence the LM statistic defined by (4.5) can be written as

$$LM_{iT}(p_i, \boldsymbol{\rho}_i) = \frac{T \boldsymbol{\xi}'_i \mathbf{P}_i^* \boldsymbol{\xi}_i}{\boldsymbol{\xi}'_i \mathbf{M}_{Q_i} \boldsymbol{\xi}_i}. \quad (\text{A.41})$$

It is easily seen that

$$\boldsymbol{\xi}'_i \mathbf{M}_{Q_i} \boldsymbol{\xi}_i = \boldsymbol{\varepsilon}'_i \mathbf{M}_{Q_i} \boldsymbol{\varepsilon}_i + A_{iT}^*, \quad \boldsymbol{\xi}'_i \mathbf{P}_i^* \boldsymbol{\xi}_i = \boldsymbol{\varepsilon}'_i \mathbf{P}_i^* \boldsymbol{\varepsilon}_i + A_{iT}^*,$$

where $A_{iT}^* = \beta_i^2 \mathbf{y}'_{i,-1} \mathbf{M}_{Q_i} \mathbf{y}_{i,-1} + 2\beta_i \boldsymbol{\varepsilon}'_i \mathbf{M}_{Q_i} \mathbf{y}_{i,-1}$. Using these in (A.41), we now have

$$LM_{iT}(p_i, \boldsymbol{\rho}_i) - d_{iT}(\beta_i) = \frac{TA_{iT}^* (\boldsymbol{\varepsilon}'_i \mathbf{M}_{Q_i} \boldsymbol{\varepsilon}_i - \boldsymbol{\varepsilon}'_i \mathbf{P}_i^* \boldsymbol{\varepsilon}_i)}{(\boldsymbol{\varepsilon}'_i \mathbf{M}_{Q_i} \boldsymbol{\varepsilon}_i) (\boldsymbol{\xi}'_i \mathbf{M}_{Q_i} \boldsymbol{\xi}_i)} = \frac{TA_{iT}^*}{(\boldsymbol{\xi}'_i \mathbf{M}_{Q_i} \boldsymbol{\xi}_i)} [1 - T^{-1} d_{iT}(\beta_i)] = T(C_{1iT} + C_{2iT}), \quad (\text{A.42})$$

where

$$d_{iT}(\beta_i) = \frac{T \boldsymbol{\varepsilon}'_i \mathbf{P}_i^* (\beta_i) \boldsymbol{\varepsilon}_i}{\boldsymbol{\varepsilon}'_i \mathbf{M}_{Q_i} \boldsymbol{\varepsilon}_i}, \quad C_{1iT} = \frac{\beta_i^2 (T^{-1} \mathbf{y}'_{i,-1} \mathbf{M}_{Q_i} \mathbf{y}_{i,-1})}{(T^{-1} \boldsymbol{\varepsilon}'_i \boldsymbol{\varepsilon}_i) + \beta_i^2 (T^{-1} \mathbf{y}'_{i,-1} \mathbf{M}_{Q_i} \mathbf{y}_{i,-1})},$$

$$\begin{aligned} C_{2iT} = & \frac{1}{\sqrt{T}} \frac{2\beta_i (T^{-1} \boldsymbol{\varepsilon}'_i \boldsymbol{\varepsilon}_i) (T^{-\frac{1}{2}} \boldsymbol{\varepsilon}'_i \mathbf{M}_{Q_i} \mathbf{y}_{i,-1})}{(T^{-1} \boldsymbol{\xi}'_i \mathbf{M}_{Q_i} \boldsymbol{\xi}_i) [(T^{-1} \boldsymbol{\varepsilon}'_i \boldsymbol{\varepsilon}_i) + \beta_i^2 (T^{-1} \mathbf{y}'_{i,-1} \mathbf{M}_{Q_i} \mathbf{y}_{i,-1})]} - \frac{1}{T} \frac{T^{-1} A_{iT}^* d_{iT}(\beta_i)}{(T^{-1} \boldsymbol{\xi}'_i \mathbf{M}_{Q_i} \boldsymbol{\xi}_i)} \\ & + \frac{1}{T} \frac{T^{-1} \beta_i^2 (T^{-1} \mathbf{y}'_{i,-1} \mathbf{M}_{Q_i} \mathbf{y}_{i,-1}) (T^{-\frac{1}{2}} \boldsymbol{\varepsilon}'_i \mathbf{Q}_i) (T^{-1} \mathbf{Q}'_i \mathbf{Q}_i)^{-1} (T^{-\frac{1}{2}} \mathbf{Q}'_i \boldsymbol{\varepsilon}_i)}{(T^{-1} \boldsymbol{\xi}'_i \mathbf{M}_{Q_i} \boldsymbol{\xi}_i) [(T^{-1} \boldsymbol{\varepsilon}'_i \boldsymbol{\varepsilon}_i) + \beta_i^2 (T^{-1} \mathbf{y}'_{i,-1} \mathbf{M}_{Q_i} \mathbf{y}_{i,-1})]}. \end{aligned}$$

[A.7]

Consider now the terms involving C_{1iT} and C_{2iT} . It is easily seen that C_{1iT} has second-order moments for a finite T and as $T \rightarrow \infty$,

$$C_{1iT} \xrightarrow{p} \frac{\beta_i^2 w_i^2}{\sigma_i^2 + \beta_i^2 w_i^2},$$

where $w_i^2 = \text{plim}_{T \rightarrow \infty} (T^{-1} \mathbf{y}'_{i,-1} \mathbf{M}_{Q_i} \mathbf{y}_{i,-1}) < \infty$. Since $LM_{iT}(p_i, \boldsymbol{\rho}_i)$ and $d_{iT}(\beta_i)$ have second-order moments, it also follows that the second-order moment of C_{2iT} exists. Furthermore, $d_{iT}(\beta_i) = O_p(1)$ and $C_{2iT} = O_p(T^{-1/2})$.

Averaging (A.42) across i ,

$$\begin{aligned} \Gamma_{\overline{LM}} &= \frac{\sum_{i=1}^N \{d_{iT}(\beta_i) - E(\eta_T)\}}{\sqrt{N \text{Var}(\eta_T)}} + \frac{\sum_{i=1}^N \{LM_{iT}(p_i, \boldsymbol{\rho}_i) - d_{iT}(\beta_i)\}}{\sqrt{N \text{Var}(\eta_T)}} \\ &= \frac{\sum_{i=1}^N \{d_{iT}(\beta_i) - E(\eta_T)\}}{\sqrt{N \text{Var}(\eta_T)}} + \frac{T\sqrt{N}}{\sqrt{\text{Var}(\eta_T)}} \left\{ \frac{1}{N} \sum_{i=1}^N C_{1iT} + \frac{1}{N} \sum_{i=1}^N C_{2iT} \right\}. \end{aligned}$$

Dividing $\Gamma_{\overline{LM}}$ by $T\sqrt{N}$, we have

$$\frac{1}{T\sqrt{N}} \Gamma_{\overline{LM}} = \frac{1}{TN} \sum_{i=1}^N \frac{d_{iT}(\beta_i) - E(\eta_T)}{\sqrt{\text{Var}(\eta_T)}} + \frac{1}{\sqrt{\text{Var}(\eta_T)}} \left\{ \frac{1}{N} \sum_{i=1}^N C_{1iT} + \frac{1}{N} \sum_{i=1}^N C_{2iT} \right\}. \quad (\text{A.43})$$

Since second-order moments of d_{iT} , C_{1iT} and C_{2iT} exist for all T , then by the strong law of large numbers, as $T \rightarrow \infty$, we have (recall that d_{iT} , C_{1iT} and C_{2iT} are also independently distributed across i):

$$\begin{aligned} \frac{1}{N} \sum_{i=1}^N \frac{d_{iT}(\beta_i) - E(\eta_T)}{\sqrt{\text{Var}(\eta_T)}} - \frac{1}{N} \sum_{i=1}^N \frac{E[d_{iT}(\beta_i)] - E(\eta_T)}{\sqrt{\text{Var}(\eta_T)}} &\xrightarrow{a.s.} 0, \\ \frac{1}{N} \sum_{i=1}^N C_{1iT} - \frac{1}{N} \sum_{i=1}^N E(C_{1iT}) &\xrightarrow{a.s.} 0, \quad \frac{1}{N} \sum_{i=1}^N C_{2iT} - \frac{1}{N} \sum_{i=1}^N E(C_{2iT}) \xrightarrow{a.s.} 0. \end{aligned}$$

But, using the above results,

$$\lim_{T \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N \frac{E[d_{iT}(\beta_i)] - E(\eta_T)}{\sqrt{\text{Var}(\eta_T)}} < \infty, \quad \lim_{T \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N E(C_{2iT}) = 0,$$

and finally¹⁴

$$\lim_{T \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N C_{1iT} = \lim_{N \rightarrow \infty} \frac{N_1}{N} \times \lim_{N_1 \rightarrow \infty} \left\{ \frac{1}{N_1} \sum_{i=1}^{N_1} \frac{\beta_i^2 w_i^2}{\sigma_i^2 + \beta_i^2 w_i^2} \right\}.$$

Using the above results in (A.43),

$$\text{plim}_{T \rightarrow \infty} \left(\frac{1}{T\sqrt{N}} \Gamma_{\overline{LM}} \right) = \frac{\delta_1}{\sqrt{\text{Var}(\eta_T)}} \lim_{N_1 \rightarrow \infty} \left\{ \frac{1}{N_1} \sum_{i=1}^{N_1} \frac{\beta_i^2 w_i^2}{\sigma_i^2 + \beta_i^2 w_i^2} \right\},$$

where $\delta_1 = \lim_{N \rightarrow \infty} (N_1/N) > 0$. This establishes the consistency of the test for $0 < \sigma_i^2$, $w_i^2 < \infty$ and $\beta_i^2 > 0$, $i = 1, 2, \dots, N_1$. ■

¹⁴Recall that $\beta_i = 0$ for $i = N_1 + 1, N_1 + 2, \dots, N$.

A.5 Evaluation of $E(K_{iT})$

Let $\varphi_i(\ell, k)$, $\ell, k = 1, 2, \dots, p_i$, be (ℓ, k) -th element of $[E(T^{-1}\mathbf{W}_i'\mathbf{M}_\tau\mathbf{W}_i)]^{-1}$, then

$$K_{iT} = \frac{\sum_{\ell=1}^{p_i} \sum_{k=1}^{p_i} \varphi_i(\ell, k) (T^{-1}\boldsymbol{\varepsilon}_i'\mathbf{M}_\tau\mathbf{s}_{i,-1}) (T^{-1/2}\boldsymbol{\varepsilon}_i'\mathbf{M}_\tau\Delta\mathbf{y}_{i,-\ell}) (T^{-1}\Delta\mathbf{y}_{i,-k}'\mathbf{M}_\tau\mathbf{s}_{i,-1})}{E(T^{-1}\boldsymbol{\varepsilon}_i'\mathbf{M}_\tau\boldsymbol{\varepsilon}_i) E(T^{-2}\mathbf{s}_{i,-1}'\mathbf{M}_\tau\mathbf{s}_{i,-1})},$$

Since $\varphi_i(\ell, k) = O(1)$, $E(T^{-1}\boldsymbol{\varepsilon}_i'\mathbf{M}_\tau\boldsymbol{\varepsilon}_i) = O(1)$, $E(T^{-2}\mathbf{s}_{i,-1}'\mathbf{M}_\tau\mathbf{s}_{i,-1}) = O(1)$, so as to establish that $E(K_{iT}) = O(T^{-1/2})$ it is sufficient to show that for all $\ell, k = 1, 2, \dots, p_i$,

$$E(\boldsymbol{\varepsilon}_i'\mathbf{M}_\tau\mathbf{s}_{i,-1}) (\boldsymbol{\varepsilon}_i'\mathbf{M}_\tau\Delta\mathbf{y}_{i,-\ell}) (\Delta\mathbf{y}_{i,-k}'\mathbf{M}_\tau\mathbf{s}_{i,-1}) = O(T^2).$$

We now show that this is in fact true for $\ell = k = 1$.

Under $\beta_i = 0$, Δy_{it} , $t = 0, 1, 2, \dots, T-1$, is given by (A.22), which can be written in the matrix form as (we have dropped the subscript i to simplify the exposition)

$$\mathbf{P}\Delta\mathbf{y}_{-1} = \mathbf{\Lambda}\Delta\mathbf{y}^0 + \boldsymbol{\varepsilon}_{-1}, \quad (\text{A.44})$$

where $\Delta\mathbf{y}_{-1} = (\Delta y_0, \Delta y_1, \dots, \Delta y_{T-1})$, $\Delta\mathbf{y}^0 = (\Delta y_{-1}, \Delta y_{-2}, \dots, \Delta y_{-p})'$, $\boldsymbol{\varepsilon}_{-1} = (\varepsilon_0, \varepsilon_1, \dots, \varepsilon_{T-1})'$,

$$\mathbf{P}_{T \times T} = \begin{bmatrix} 1 & 0 & 0 & \cdots & \cdots & 0 \\ -\rho_1 & 1 & 0 & \cdots & \cdots & 0 \\ -\rho_2 & -\rho_1 & 1 & \cdots & \cdots & 0 \\ & & & & \vdots & \\ 0 & 0 & 0 & \cdots & -\rho_p & \cdots & 1 \end{bmatrix}, \quad \mathbf{\Lambda}_{T \times p} = \begin{bmatrix} \rho_1 & \rho_2 & \cdots & \rho_{p-1} & \rho_p \\ \rho_2 & \rho_3 & \cdots & \rho_p & 0 \\ \rho_3 & \rho_4 & \cdots & 0 & 0 \\ & & & \vdots & \\ \rho_p & 0 & \cdots & 0 & 1 \\ & & & \mathbf{0} & \end{bmatrix}$$

Define the inverse of $\mathbf{P}$ by $\boldsymbol{\Psi}$, then it is easily seen that $\boldsymbol{\Psi}$ is a lower triangular matrix,

$$\boldsymbol{\Psi}_{T \times T} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ \psi_1 & 1 & 0 & \cdots & 0 \\ \psi_2 & \psi_1 & 1 & \cdots & 0 \\ & & & \vdots & \\ \psi_{T-1} & \psi_{T-2} & \psi_{T-3} & \cdots & 1 \end{bmatrix},$$

where

$$\psi_j = \rho_1\psi_{j-1} + \rho_2\psi_{j-2} + \cdots + \rho_p\psi_{j-p}, \quad j = 1, 2, \dots, \quad (\text{A.45})$$

with $\psi_0 = 1$ and $\psi_j = 0$ for $j < 0$. Since ψ_j 's are generated by a stable difference equation (A.45), hence $\sum_{j=1}^T |\psi_{j-1}| = O(1)$. Notice also that as $j \rightarrow \infty$, $\psi_j \rightarrow 0$ at a geometric rate.

Pre-multiplying (A.44) by $\boldsymbol{\Psi}$, we now have

$$\Delta\mathbf{y}_{-1} = \boldsymbol{\Psi}\boldsymbol{\varepsilon}_{-1} + \boldsymbol{\Psi}\mathbf{\Lambda}\Delta\mathbf{y}^0. \quad (\text{A.46})$$

Notice that $E(\Delta\mathbf{y}^0) = \mathbf{0}$, $\text{Var}(\Delta\mathbf{y}^0) = \boldsymbol{\Sigma}^{00}$, where $\boldsymbol{\Sigma}^{00}$ is the $p \times p$ positive definite covariance matrix of a stationary process. Using $\boldsymbol{\varepsilon}_{-1} = \varepsilon_0(1, 0, \dots, 0)' + \mathbf{L}\boldsymbol{\varepsilon}$ in (A.46) where $\mathbf{L}$ is defined by (3.6), then

$$\Delta\mathbf{y}_{-1} = \boldsymbol{\Psi}\mathbf{L}\boldsymbol{\varepsilon} + \mathbf{d}, \quad (\text{A.47})$$

where $\mathbf{d} = \psi_1\varepsilon_0 + \boldsymbol{\Psi}\mathbf{\Lambda}\Delta\mathbf{y}^0$ with $\psi_1 = (1, \psi_1, \psi_2, \dots, \psi_{T-1})'$. Notice that ε_0 and $\Delta\mathbf{y}^0$ are independently distributed, and therefore, $E(\mathbf{d}) = \mathbf{0}$, $\text{Var}(\mathbf{d}) = \sigma^2\psi_1\psi_1' + \boldsymbol{\Psi}\mathbf{\Lambda}\boldsymbol{\Sigma}^{00}\mathbf{\Lambda}'\boldsymbol{\Psi}'$. It is now easily seen that the sum of the elements in each column of $\text{Var}(\mathbf{d})$ is bounded in T . $\boldsymbol{\Lambda}\boldsymbol{\Sigma}^{00}\mathbf{\Lambda}'$ does not vary with T , and the elements in ψ_1 tend to zero at a geometric rate.

Using the above results, noting that $\mathbf{s}_{-1} = \mathbf{H}\mathbf{L}\boldsymbol{\varepsilon}$ where $\mathbf{H}$ is defined by (3.6), and defining

$$\mathbf{A}_1 = \mathbf{H}\mathbf{L}, \quad \tilde{\mathbf{A}}_1 = \mathbf{M}_\tau\mathbf{A}_1, \quad \mathbf{A}_2 = \boldsymbol{\Psi}\mathbf{L}, \quad \tilde{\mathbf{A}}_2 = \mathbf{M}_\tau\mathbf{A}_2, \quad \tilde{\mathbf{A}}_3 = \tilde{\mathbf{A}}_2'\tilde{\mathbf{A}}_1, \quad (\text{A.48})$$

[A.9]

we now have¹⁵

$$\begin{aligned}
& E(\varepsilon' \mathbf{M}_\tau \mathbf{s}_{-1}) (\varepsilon' \mathbf{M}_\tau \Delta \mathbf{y}_{-1}) (\Delta \mathbf{y}_{-1}' \mathbf{M}_\tau \mathbf{s}_{-1}) \\
&= E(\varepsilon' \tilde{\mathbf{A}}_1 \varepsilon) (\varepsilon' \mathbf{M}_\tau \mathbf{d}) (\mathbf{d}' \tilde{\mathbf{A}}_1 \varepsilon) + E(\varepsilon' \tilde{\mathbf{A}}_1 \varepsilon) (\varepsilon' \tilde{\mathbf{A}}_2 \varepsilon) (\varepsilon' \tilde{\mathbf{A}}_3 \varepsilon) \\
&\quad + E(\varepsilon' \tilde{\mathbf{A}}_1 \varepsilon) \left[(\varepsilon' \mathbf{M}_\tau \mathbf{d}) (\varepsilon' \tilde{\mathbf{A}}_3 \varepsilon) + (\varepsilon' \tilde{\mathbf{A}}_2 \varepsilon) (\mathbf{d}' \tilde{\mathbf{A}}_1 \varepsilon) \right].
\end{aligned} \tag{A.49}$$

Notice that the expectation of the last terms will become zero as $\mathbf{d}$ is independent of ε .

It remains to evaluate the first and the second terms. Since the direct evaluation is quite tedious and complicated, we will use the following result:

$$|tr \mathbf{A}_1 \mathbf{B}| \leq \tau'_T / \mathbf{B} / \tau_T, \quad |tr \mathbf{A}_1' \mathbf{B}| \leq \tau'_T / \mathbf{B} / \tau_T, \tag{A.50}$$

where $\mathbf{A}_1$ is defined in (A.48), $\mathbf{B}$ is any matrix conformable to $\mathbf{A}_1$, and $/\mathbf{B}/$ denotes the matrix $\mathbf{B}$ with all its elements in absolute value. It is also straightforward to show that

$$tr(\mathbf{A}_1) = 0, \quad tr(\mathbf{A}_2) = 0, \quad \tau'_T \mathbf{A}_1 \tau_T = O(T^2), \quad \tau'_T \mathbf{A}_2 \tau_T = O(T), \tag{A.51}$$

$$tr(\tilde{\mathbf{A}}_1) = O(T), \quad tr(\tilde{\mathbf{A}}_2) = O(1), \tag{A.52}$$

$$tr(\mathbf{A}_1 \mathbf{A}_1) = O(T^2), \quad tr(\mathbf{A}_1 \mathbf{A}_2) = O(T), \quad tr(\mathbf{A}_2 \mathbf{A}_2) = O(T). \tag{A.53}$$

Consider the first term. Using the independence between $\mathbf{d}$ and ε , and applying Lemma A.4, then

$$E(\varepsilon' \tilde{\mathbf{A}}_1 \varepsilon) (\varepsilon' \mathbf{M}_\tau \mathbf{d} \mathbf{d}' \tilde{\mathbf{A}}_1 \varepsilon) = tr(\tilde{\mathbf{A}}_1) tr[\mathbf{M}_\tau E(\mathbf{d} \mathbf{d}') \tilde{\mathbf{A}}_1] + tr[(\tilde{\mathbf{A}}_1 + \tilde{\mathbf{A}}_1') \mathbf{M}_\tau E(\mathbf{d} \mathbf{d}') \tilde{\mathbf{A}}_1].$$

Next, note that

$$\begin{aligned}
\left| tr[\mathbf{M}_\tau E(\mathbf{d} \mathbf{d}') \tilde{\mathbf{A}}_1] \right| &\leq |d_{\max}| \left| tr(\tilde{\mathbf{A}}_1 \mathbf{M}_\tau) \right|, \\
\left| tr[(\tilde{\mathbf{A}}_1 + \tilde{\mathbf{A}}_1') \mathbf{M}_\tau E(\mathbf{d} \mathbf{d}') \tilde{\mathbf{A}}_1] \right| &\leq |d_{\max}| \left| tr[(\tilde{\mathbf{A}}_1 + \tilde{\mathbf{A}}_1') \tilde{\mathbf{A}}_1] \right|,
\end{aligned}$$

where $d_{\max}$ is the maximum eigenvalue of $E(\mathbf{d} \mathbf{d}')$. Since $d_{\max} = O(1)$, $tr(\tilde{\mathbf{A}}_1 \mathbf{M}_\tau) = tr(\tilde{\mathbf{A}}_1) = O(T)$,

$$\begin{aligned}
\left| tr(\tilde{\mathbf{A}}_1 \tilde{\mathbf{A}}_1) \right| &\leq |tr(\mathbf{A}_1 \mathbf{A}_1)| + 2 \left| \frac{1}{T} \tau'_T \mathbf{A}_1 \mathbf{A}_1 \tau_T \right| + \left| \frac{1}{T} \tau'_T \mathbf{A}_1 \tau_T \right|^2 \\
&\leq |tr(\mathbf{A}_1 \mathbf{A}_1)| + 2 |\tau'_T \mathbf{A}_1 \tau_T| + \left| \frac{1}{T} \tau'_T \mathbf{A}_1 \tau_T \right|^2 = O(T^2),
\end{aligned}$$

and similarly, $tr(\tilde{\mathbf{A}}_1' \tilde{\mathbf{A}}_1) = O(T^2)$, hence

$$E(\varepsilon' \tilde{\mathbf{A}}_1 \varepsilon) (\varepsilon' \mathbf{M}_\tau \mathbf{d} \mathbf{d}' \tilde{\mathbf{A}}_1 \varepsilon) = O(T^2). \tag{A.54}$$

Next, consider the second term in (A.49). Using (A.50) and after some algebra, we obtain¹⁶

$$tr(\tilde{\mathbf{A}}_1 \tilde{\mathbf{A}}_2 + \tilde{\mathbf{A}}_1' \tilde{\mathbf{A}}_2) = O(T), \quad tr(\tilde{\mathbf{A}}_1 \tilde{\mathbf{A}}_3 + \tilde{\mathbf{A}}_1' \tilde{\mathbf{A}}_3) = O(T^2), \quad tr(\tilde{\mathbf{A}}_2 \tilde{\mathbf{A}}_3 + \tilde{\mathbf{A}}_2' \tilde{\mathbf{A}}_3) = O(T), \tag{A.55}$$

$$tr(\mathbf{A}_1 \tilde{\mathbf{A}}_2 \tilde{\mathbf{A}}_3 + \mathbf{A}_1' \tilde{\mathbf{A}}_2 \tilde{\mathbf{A}}_3 + \mathbf{A}_1 \tilde{\mathbf{A}}_2' \tilde{\mathbf{A}}_3 + \mathbf{A}_1 \tilde{\mathbf{A}}_2 \tilde{\mathbf{A}}_3') = O(T^2). \tag{A.56}$$

Applying Lemma A.4 to the second term in (A.49), and using (A.51) - (A.56), it is easily seen that

$$E(\varepsilon' \tilde{\mathbf{A}}_1 \varepsilon) (\varepsilon' \tilde{\mathbf{A}}_2 \varepsilon) (\varepsilon' \tilde{\mathbf{A}}_3 \varepsilon) = O(T^2). \tag{A.57}$$

Hence, using (A.54) and (A.57), then

$$E(\varepsilon' \mathbf{M}_\tau \mathbf{s}_{-1}) (\varepsilon' \mathbf{M}_\tau \Delta \mathbf{y}_{-1}) (\Delta \mathbf{y}_{-1}' \mathbf{M}_\tau \mathbf{s}_{-1}) = O(T^2).$$

We can obtain the same result for other values of ℓ and k along similar lines. Therefore, $E(K_{iT}) = O(T^{-1/2})$. ■

¹⁵Without loss of generality we assume that $\varepsilon \sim N(\mathbf{0}, \mathbf{I}_T)$.

¹⁶Detailed derivations are available upon request.

A.6 Proof of Theorem 4.2

The theorem is proved if under the null hypothesis ($\beta_i = 0, \forall i$)

$$\sqrt{N} \left(\widetilde{LM}_{NT} - \bar{\eta}_{NT} \right) \xrightarrow{p} 0,$$

as both N and $T \rightarrow \infty$, where $\widetilde{LM}_{NT} = N^{-1} \sum_{i=1}^N \widetilde{LM}_{iT}$, $\bar{\eta}_{NT} = N^{-1} \sum_{i=1}^N \eta_{iT}$, and $\widetilde{LM}_{iT}$ and η_{iT} are defined, respectively, by (4.12) and (3.5). Since the common time-specific effects, θ_t , are differenced away in the de-meaned regressions (4.10), then under the null hypothesis $\widetilde{LM}_{iT}$ can be written as

$$\widetilde{LM}_{iT} = T \frac{[(\varepsilon_i - \bar{\varepsilon})' \mathbf{M}_\tau (\mathbf{s}_{i,-1} - \bar{\mathbf{s}}_{-1})]^2}{[(\varepsilon_i - \bar{\varepsilon})' \mathbf{M}_\tau (\varepsilon_i - \bar{\varepsilon})] [(\mathbf{s}_{i,-1} - \bar{\mathbf{s}}_{-1})' \mathbf{M}_\tau (\mathbf{s}_{i,-1} - \bar{\mathbf{s}}_{-1})]},$$

where

$$\bar{\mathbf{s}}_{-1} = (\bar{s}_0, \bar{s}_1, \dots, \bar{s}_{T-1})', \quad \bar{s}_t = \frac{1}{N} \sum_{i=1}^N s_{it}, \quad \bar{\varepsilon} = (\bar{\varepsilon}_1, \bar{\varepsilon}_2, \dots, \bar{\varepsilon}_T)', \quad \bar{\varepsilon}_t = \frac{1}{N} \sum_{i=1}^N \varepsilon_{it}.$$

After some algebraic manipulations we have¹⁷

$$\widetilde{LM}_{iT} - \eta_{iT} = -a_i + (\eta_{iT} - a_i) \left(\frac{b_i}{1 - b_i} + \frac{c_i}{1 - c_i} + \frac{b_i c_i}{(1 - b_i)(1 - c_i)} \right), \quad (\text{A.58})$$

where

$$a_i = T \frac{2\Psi_i (\varepsilon_i' \mathbf{M}_\tau \mathbf{s}_{i,-1}) + \Psi_i^2}{(\varepsilon_i' \mathbf{M}_\tau \varepsilon_i) (\mathbf{s}_{i,-1}' \mathbf{M}_\tau \mathbf{s}_{i,-1})} \quad (\text{A.59})$$

$$\Psi_i = \bar{\varepsilon}' \mathbf{M}_\tau (\mathbf{s}_{i,-1} - \bar{\mathbf{s}}_{-1}) + \varepsilon_i' \mathbf{M}_\tau \bar{\mathbf{s}}_{-1}, \quad (\text{A.60})$$

$$b_i = \frac{2\bar{\varepsilon}' \mathbf{M}_\tau \varepsilon_i - \bar{\varepsilon}' \mathbf{M}_\tau \bar{\varepsilon}}{(\varepsilon_i' \mathbf{M}_\tau \varepsilon_i)}, \quad \text{and} \quad c_i = \frac{2\bar{\mathbf{s}}_{-1}' \mathbf{M}_\tau \mathbf{s}_{i,-1} - \bar{\mathbf{s}}_{-1}' \mathbf{M}_\tau \bar{\mathbf{s}}_{-1}}{(\mathbf{s}_{i,-1}' \mathbf{M}_\tau \mathbf{s}_{i,-1})}. \quad (\text{A.61})$$

Also $\mathbf{s}_{i,-1}' \mathbf{M}_\tau \mathbf{s}_{i,-1} = \varepsilon_i' \mathbf{L}' \mathbf{H}' \mathbf{M}_\tau \mathbf{H} \mathbf{L} \varepsilon_i$, where the $T \times T$ matrices $\mathbf{H}$ and $\mathbf{L}$ are defined by (3.6). Let $\mathbf{A} = \mathbf{L}' \mathbf{H}' \mathbf{M}_\tau \mathbf{H} \mathbf{L} = \tilde{\mathbf{A}}_1' \tilde{\mathbf{A}}_1$, and $\tilde{\mathbf{A}}_1 = \mathbf{M}_\tau \mathbf{H} \mathbf{L}$, which is already defined by (A.48). It is clear that $\mathbf{M}_\tau$ and $\mathbf{A}$ are semi-positive definite matrices of rank $T - 1$, and by Lemma A.5 and Corollary A.1 the inverses of $(\varepsilon_i' \mathbf{M}_\tau \varepsilon_i)$, $(\mathbf{s}_{i,-1}' \mathbf{M}_\tau \mathbf{s}_{i,-1})$ and $(\varepsilon_i' \mathbf{M}_\tau \varepsilon_i) (\mathbf{s}_{i,-1}' \mathbf{M}_\tau \mathbf{s}_{i,-1})$ have at least second-order moments if $T > 9$. Therefore, by Corollary A.1 and Lemma A.6 the second-order moment of a_i also exists for $T > 9$. Finally, since $\widetilde{LM}_{iT}$ and η_{iT} have second-order moments, then so will d_i defined by

$$d_i = \widetilde{LM}_{iT} - \eta_{iT} + a_i = (\eta_{iT} - a_i) \left(\frac{b_i}{1 - b_i} + \frac{c_i}{1 - c_i} + \frac{b_i c_i}{(1 - b_i)(1 - c_i)} \right). \quad (\text{A.62})$$

Given the above results we can write

$$\sqrt{N} \left(\widetilde{LM}_{NT} - \bar{\eta}_{NT} \right) = -N^{-1/2} \sum_{i=1}^N a_i + N^{-1} \sum_{i=1}^N \sqrt{N} d_i. \quad (\text{A.63})$$

However, note that a_i and d_i are neither identically nor independently distributed across i , and standard laws of large numbers for cross-sectional observations are not applicable here. A more direct approach is required. Consider first the second term in (A.63). Since for a large enough value of T (namely $T > 9$) the second-order moment of d_i (defined by (A.62)) exists, then Lemma A.7 is directly applicable. Using (A.2), (A.3) and (A.5) it readily follows that $\eta_{iT} = O_p(1)$. Similarly, since

$$a_i = \frac{2(T^{-1} \Psi_i) (T^{-1} \varepsilon_i' \mathbf{M}_\tau \mathbf{s}_{i,-1}) + (T^{-1} \Psi_i)^2}{(T^{-1} \varepsilon_i' \mathbf{M}_\tau \varepsilon_i) (T^{-1} \mathbf{s}_{i,-1}' \mathbf{M}_\tau \mathbf{s}_{i,-1})}, \quad (\text{A.64})$$

¹⁷ a_i , b_i and c_i also depend on N and T , but for notational simplicity we are not showing these additional subscripts.

then the order of a_i is the same as that of $T^{-1}\Psi_i$, but it is easily seen that $T^{-1}\Psi_i = O_p(N^{-1/2})$. Hence, the orders of $E(\sqrt{N}d_i)$ and $Var(\sqrt{N}d_i)$ are determined by those of $E(\sqrt{N}b_i)$, $E(\sqrt{N}c_i)$, $Var(\sqrt{N}b_i)$ and $Var(\sqrt{N}c_i)$.

Using $\bar{\epsilon} = (\frac{1}{N})\epsilon_i + (\frac{N-1}{N})\bar{\epsilon}_{-i}$, where $\bar{\epsilon}_{-i} = \frac{1}{N-1} \sum_{j \neq i} \epsilon_j$, in (A.61) we have¹⁸

$$b_i = \left(\frac{2}{N} - \frac{1}{N^2} \right) + 2 \left(1 - \frac{1}{N} \right)^2 \left(\frac{\epsilon'_i \mathbf{M}_\tau \bar{\epsilon}_{-i}}{\epsilon'_i \mathbf{M}_\tau \epsilon_i} \right) - \left(1 - \frac{1}{N} \right)^2 \left(\frac{\bar{\epsilon}'_{-i} \mathbf{M}_\tau \bar{\epsilon}_{-i}}{\epsilon'_i \mathbf{M}_\tau \epsilon_i} \right). \quad (\text{A.65})$$

Since ϵ_i and $\bar{\epsilon}_{-i}$ are independently distributed it is easily seen that (for $T > 3$)

$$E(b_i) = \left(\frac{2}{N} - \frac{1}{N^2} \right) - \left(1 - \frac{1}{N} \right)^2 \left(1 - \frac{1}{T} \right) \left(\frac{\bar{\sigma}_{-i}^2}{N-1} \right) E \left(\frac{1}{T^{-1} \epsilon'_i \mathbf{M}_\tau \epsilon_i} \right),$$

where $\bar{\sigma}_{-i}^2 = \frac{1}{N-1} \sum_{j \neq i} \sigma_j^2 = O(1)$. Hence, $E(\sqrt{N}b_i) = O(N^{-1/2})$. Also for $T > 9$,

$$Var \left(\frac{\sqrt{N} \epsilon'_i \mathbf{M}_\tau \bar{\epsilon}_{-i}}{\epsilon'_i \mathbf{M}_\tau \epsilon_i} \right) = T^{-1} \bar{\sigma}_{-i}^2 \left(\frac{N}{N-1} \right) E \left(\frac{1}{T^{-1} \epsilon'_i \mathbf{M}_\tau \epsilon_i} \right) = O(T^{-1}),$$

$$Var \left(\frac{\sqrt{N} \bar{\epsilon}'_{-i} \mathbf{M}_\tau \bar{\epsilon}_{-i}}{\epsilon'_i \mathbf{M}_\tau \epsilon_i} \right) = \left[\frac{N \bar{\sigma}_{-i}^4}{(N-1)^2} \right] \left\{ \left(1 - \frac{1}{T^2} \right) E \left[(T^{-1} \epsilon'_i \mathbf{M}_\tau \epsilon_i)^{-2} \right] - \left(1 - \frac{1}{T} \right)^2 E \left[(T^{-1} \epsilon'_i \mathbf{M}_\tau \epsilon_i)^{-1} \right]^2 \right\}.$$

Therefore, $Var(\sqrt{N}b_i) = O(T^{-1}) + O(N^{-1}) + O(T^{-1/2}N^{-1/2})$.¹⁹ From these results it now readily follows that $\sqrt{N}b_i \xrightarrow{p} 0$, as both N and T tend to infinity.

To establish the orders of the mean and the variance of $\sqrt{N}c_i$, from (A.61) we have

$$c_i = \left(\frac{2}{N} - \frac{1}{N^2} \right) + \left(1 - \frac{1}{N} \right)^2 \left(\frac{\epsilon'_i \mathbf{A} \bar{\epsilon}_{-i}}{\epsilon'_i \mathbf{A} \epsilon_i} \right) - \left(1 - \frac{1}{N} \right)^2 \left(\frac{\bar{\epsilon}'_{-i} \mathbf{A} \bar{\epsilon}_{-i}}{\epsilon'_i \mathbf{A} \epsilon_i} \right),$$

where $\mathbf{A} = \tilde{\mathbf{A}}_1' \tilde{\mathbf{A}}_1$, and $\tilde{\mathbf{A}}_1 = \mathbf{M}_\tau \mathbf{H} \mathbf{L}$. Using an analogous line of reasoning as above and noting that $tr(\mathbf{A}) = O(T^2)$, $tr(\mathbf{A}^2) = O(T^3)$, it is also easily established that $E(\sqrt{N}c_i) = O(N^{-1/2})$, and $Var(\sqrt{N}c_i) = O(T^{-1}) + O(N^{-1}) + O(T^{-1/2}N^{-1/2})$, which also implies that $\sqrt{N}c_i \xrightarrow{p} 0$, as both N and T tend to infinity. In view of the above results it now readily follows that $\sqrt{N}d_i \xrightarrow{p} 0$ as required, and by Lemma A.7 $N^{-1} \sum_{i=1}^N \sqrt{N}d_i \xrightarrow{p} 0$, as both N and T tend to infinity.

Consider now the first term in (A.63). Noting that $T^{-1}\Psi_i = O_p(N^{-1/2})$, we have

$$a_i = \frac{2(T^{-1}\Psi_i)(T^{-1}\epsilon'_i \mathbf{M}_\tau \mathbf{s}_{i,-1})}{(T^{-1}\epsilon'_i \mathbf{M}_\tau \epsilon_i)(T^{-2}\mathbf{s}'_{i,-1} \mathbf{M}_\tau \mathbf{s}_{i,-1})} + O_p(N^{-1}).$$

Therefore, the proof of the theorem will be complete if we show that

$$N^{-1/2} \sum_{i=1}^N J_i = o_p(1), \quad (\text{A.66})$$

where (using (A.60))

$$\begin{aligned} J_i &= \frac{(T^{-1}\Psi_i)(T^{-1}\epsilon'_i \mathbf{M}_\tau \mathbf{s}_{i,-1})}{(T^{-1}\epsilon'_i \mathbf{M}_\tau \epsilon_i)(T^{-2}\mathbf{s}'_{i,-1} \mathbf{M}_\tau \mathbf{s}_{i,-1})} \\ &= \left(1 - \frac{1}{N} \right)^2 J_{1i} + \left(1 - \frac{1}{N} \right)^2 J_{2i} - (T^{-1}\bar{\epsilon}' \mathbf{M}_\tau \bar{\mathbf{s}}_{-1}) J_{3i} + \left(\frac{2}{N} - \frac{1}{N^2} \right) \eta_{iT}, \end{aligned} \quad (\text{A.67})$$

¹⁸ $\sum_{j \neq i}$ denotes summation over $j = 1, \dots, N$, except for $j = i$.

¹⁹In writing down this last expression we have made use of the Cauchy-Schwarz inequality applied to the covariance of the two stochastic terms in (A.65).

η_{iT} is defined as before by (3.5),

$$J_{1i} = \frac{T (\bar{\epsilon}'_{-i} \mathbf{M}_\tau \mathbf{s}_{i,-1}) (\epsilon'_i \mathbf{M}_\tau \mathbf{s}_{i,-1})}{(\epsilon'_i \mathbf{M}_\tau \epsilon_i) (\mathbf{s}'_{i,-1} \mathbf{M}_\tau \mathbf{s}_{i,-1})}, \quad J_{2i} = \frac{T (\epsilon'_i \mathbf{M}_\tau \bar{\mathbf{s}}_{-i,-1}) (\epsilon'_i \mathbf{M}_\tau \mathbf{s}_{i,-1})}{(\epsilon'_i \mathbf{M}_\tau \epsilon_i) (\mathbf{s}'_{i,-1} \mathbf{M}_\tau \mathbf{s}_{i,-1})}, \quad (A.68)$$

$$J_{3i} = \frac{(T^{-1} \epsilon'_i \mathbf{M}_\tau \mathbf{s}_{i,-1})}{(T^{-1} \epsilon'_i \mathbf{M}_\tau \epsilon_i) (T^{-2} \mathbf{s}'_{i,-1} \mathbf{M}_\tau \mathbf{s}_{i,-1})},$$

and $\bar{\mathbf{s}}_{-i,-1} = \frac{1}{N-1} \sum_{j \neq i} \mathbf{s}_{j,-1}$. Hence,

$$N^{-1/2} \sum_{i=1}^N J_i = \left(1 - \frac{1}{N}\right)^2 \left[\left(N^{-1/2} \sum_{i=1}^N J_{1i}\right) + \left(N^{-1/2} \sum_{i=1}^N J_{2i}\right) \right] - \left(\frac{\sqrt{N} \bar{\epsilon}' \mathbf{M}_\tau \bar{\mathbf{s}}_{-1}}{T}\right) \left(N^{-1} \sum_{i=1}^N J_{3i}\right) + \left(\frac{2}{N^{1/2}} - \frac{1}{N^{3/2}}\right) \frac{1}{N} \sum_{i=1}^N \eta_{iT}. \quad (A.69)$$

Since η_{iT} are independently and identically distributed across i with a finite mean then by the law of large numbers (due to Kolmogorov) $N^{-1} \sum_{i=1}^N \eta_{iT} \xrightarrow{a.s.} E(\eta_T) < \infty$, and therefore,

$$\left(\frac{2}{N^{1/2}} - \frac{1}{N^{3/2}}\right) \left(\frac{1}{N} \sum_{i=1}^N \eta_{iT}\right) \xrightarrow{a.s.} 0. \quad (A.70)$$

As far as the third term in (A.69) is concerned, since J_{3i} are independently distributed across i and have finite second-order moments for $T > 9$, then by the law of large numbers (due to Markov)

$$N^{-1} \sum_{i=1}^N J_{3i} \xrightarrow{a.s.} \lim_{N \rightarrow \infty} \left(N^{-1} \sum_{i=1}^N E(J_{3i})\right) < \infty.$$

Also it is easily seen that $\left(\frac{\sqrt{N}}{T} \bar{\epsilon}' \mathbf{M}_\tau \bar{\mathbf{s}}_{-1}\right) \xrightarrow{p} 0$, as $N \rightarrow \infty$.²⁰ Hence, as $N \rightarrow \infty$

$$\left(\frac{\sqrt{N}}{T} \bar{\epsilon}' \mathbf{M}_\tau \bar{\mathbf{s}}_{-1}\right) \left(N^{-1} \sum_{i=1}^N J_{3i}\right) \xrightarrow{p} 0. \quad (A.71)$$

Therefore, in view of (A.69) the proof will be complete if ²¹

$$Var \left(N^{-1/2} \sum_{i=1}^N J_{1i}\right) = o(1) \text{ and } Var \left(N^{-1/2} \sum_{i=1}^N J_{2i}\right) = o(1). \quad (A.72)$$

To simplify the notations let

$$J_{1i} = T \left[\frac{1}{N} \sum_{j \neq i} (\epsilon'_j \mathbf{M}_\tau \mathbf{s}_{i,-1}) q_i \right], \quad J_{2i} = T \left[\frac{1}{N} \sum_{j \neq i} (\epsilon'_i \mathbf{M}_\tau \mathbf{s}_{j,-1}) q_i \right],$$

²⁰Note that $\frac{\sqrt{N}}{T} \bar{\epsilon}' \mathbf{M}_\tau \bar{\mathbf{s}}_{-1} = \frac{\sqrt{N}}{T} \bar{\epsilon}' \tilde{\mathbf{A}}_1 \bar{\epsilon}$, where $\tilde{\mathbf{A}}_1$ is defined by (A.48), and $tr(\tilde{\mathbf{A}}_1) = O(T)$, $tr(\tilde{\mathbf{A}}_1 \tilde{\mathbf{A}}_1) = O(T^2)$, $tr(\tilde{\mathbf{A}}_1 \tilde{\mathbf{A}}_1) = O(T^2)$. Using Lemma A.4 it is now easily established that

$$E \left(\frac{\sqrt{N}}{T} \bar{\epsilon}' \tilde{\mathbf{A}}_1 \bar{\epsilon} \right) = O(N^{-1/2}), \quad Var \left(\frac{\sqrt{N}}{T} \bar{\epsilon}' \tilde{\mathbf{A}}_1 \bar{\epsilon} \right) = O(N^{-1}).$$

²¹Notice that due to the independence of $\bar{\epsilon}_{-i}$ and ϵ_i , $E(J_{1i}) = 0$ and $E(J_{2i}) = 0$. From the Cauchy-Schwarz inequality, (A.66) follows if (A.70), (A.71) and (A.72) hold.

where

$$q_i = \frac{(\boldsymbol{\varepsilon}'_i \mathbf{M}_\tau \mathbf{s}_{i,-1})}{(\mathbf{s}'_{i,-1} \mathbf{M}_\tau \mathbf{s}_{i,-1}) (\boldsymbol{\varepsilon}'_i \mathbf{M}_\tau \boldsymbol{\varepsilon}_i)} = O_p(T^{-2}).$$

We now have

$$\text{Var} \left(N^{-1/2} \sum_{i=1}^N J_{1i} \right) = \frac{1}{N} E \left(\sum_{i=1}^N J_{1i} \right)^2 = \frac{1}{N} \left(\sum_{i=1}^N E(J_{1i}^2) + \sum_{\ell \neq k} \sum_{k=1}^N E(J_{1\ell} J_{1k}) \right).$$

Since $J_{1i} = O_p(N^{-1/2})$, $\frac{1}{N} \sum_{i=1}^N E(J_{1i}^2) = O(N^{-1})$. Consider now the order of $E(J_{1\ell} J_{1k})$ for $\ell \neq k$,

$$\begin{aligned} E(J_{1\ell} J_{1k}) &= \frac{T^2}{N^2} E q_\ell q_k \left(\sum_{j \neq \ell} \boldsymbol{\varepsilon}'_j \mathbf{M}_\tau \mathbf{s}_{\ell,-1} \right) \left(\sum_{j \neq k} \boldsymbol{\varepsilon}'_j \mathbf{M}_\tau \mathbf{s}_{k,-1} \right) \\ &= \frac{T^2}{N^2} E q_\ell q_k \left[\left(\boldsymbol{\varepsilon}'_k + \sum_{j \neq \ell, k} \boldsymbol{\varepsilon}'_j \right) \mathbf{M}_\tau \mathbf{s}_{\ell,-1} \right] \left[\left(\boldsymbol{\varepsilon}'_\ell + \sum_{j \neq \ell, k} \boldsymbol{\varepsilon}'_j \right) \mathbf{M}_\tau \mathbf{s}_{k,-1} \right] \\ &= \frac{T^2}{N^2} E q_\ell q_k \left[(\boldsymbol{\varepsilon}'_k \mathbf{M}_\tau \mathbf{s}_{\ell,-1}) (\boldsymbol{\varepsilon}'_\ell \mathbf{M}_\tau \mathbf{s}_{k,-1}) + \left(\sum_{j \neq \ell, k} \boldsymbol{\varepsilon}'_j \mathbf{M}_\tau \mathbf{s}_{\ell,-1} \right) \left(\sum_{j \neq \ell, k} \boldsymbol{\varepsilon}'_j \mathbf{M}_\tau \mathbf{s}_{k,-1} \right) \right] \\ &= \frac{T^2}{N^2} E q_\ell q_k \left[(\boldsymbol{\varepsilon}'_k \mathbf{M}_\tau \mathbf{s}_{\ell,-1}) (\boldsymbol{\varepsilon}'_\ell \mathbf{M}_\tau \mathbf{s}_{k,-1}) + \sum_{j \neq \ell, k} (\boldsymbol{\varepsilon}'_j \mathbf{M}_\tau \mathbf{s}_{\ell,-1}) (\boldsymbol{\varepsilon}'_j \mathbf{M}_\tau \mathbf{s}_{k,-1}) \right]. \end{aligned}$$

The third and the fourth equality follow from the fact that $\boldsymbol{\varepsilon}_i$ and $\boldsymbol{\varepsilon}_j$ are distributed independently for $i \neq j$. The first term in the last equation is at most $O(N^{-2})$, which, after taking double summations across cross sectional units and dividing it by N , becomes $O(N^{-1})$. The second term can now be written as

$$\begin{aligned} &\frac{T^2}{N^2} \sum_{j \neq \ell, k} E q_\ell q_k (\boldsymbol{\varepsilon}'_j \mathbf{M}_\tau \mathbf{s}_{\ell,-1}) (\boldsymbol{\varepsilon}'_j \mathbf{M}_\tau \mathbf{s}_{k,-1}) \\ &= \frac{T^2}{N^2} \sum_{j \neq \ell, k} E q_\ell q_k [\mathbf{s}'_{\ell,-1} \mathbf{M}_\tau E(\boldsymbol{\varepsilon}_j \boldsymbol{\varepsilon}'_j) \mathbf{M}_\tau \mathbf{s}_{k,-1}] \\ &= \frac{T^2(N-2)}{N^2} \left(\frac{1}{N-2} \sum_{j \neq \ell, k} \sigma_j^2 \right) E(q_\ell q_k \mathbf{s}'_{\ell,-1} \mathbf{M}_\tau \mathbf{s}_{k,-1}). \end{aligned} \tag{A.73}$$

Since $E(q_\ell q_k \mathbf{s}'_{\ell,-1} \mathbf{M}_\tau \mathbf{s}_{k,-1}) = O(T^{-5/2})$ for $\ell \neq k$, the expression in (A.73) will be $O(N^{-1} T^{-1/2})$. Taking double summations over ℓ and k (with $\ell \neq k$) and dividing the result by N , it becomes $O(T^{-1/2})$. Therefore,

$$\text{Var} \left(N^{-1/2} \sum_{i=1}^N J_{1i} \right) = O(N^{-1}) + O(T^{-1/2}).$$

Similarly, for J_{2i} we have

$$E(J_{2\ell} J_{2k}) = \frac{T^2}{N^2} E q_\ell q_k \left[(\boldsymbol{\varepsilon}'_k \mathbf{M}_\tau \mathbf{s}_{\ell,-1}) (\boldsymbol{\varepsilon}'_\ell \mathbf{M}_\tau \mathbf{s}_{k,-1}) + \sum_{j \neq \ell, k} (\boldsymbol{\varepsilon}'_\ell \mathbf{M}_\tau \mathbf{s}_{j,-1}) (\boldsymbol{\varepsilon}'_k \mathbf{M}_\tau \mathbf{s}_{j,-1}) \right].$$

The first term is $O(N^{-2})$ as shown above. Noting that $\mathbf{s}_{j,-1} = \mathbf{H} \boldsymbol{\varepsilon}_{j,-1}$, the second term becomes

$$\frac{T^2}{N^2} \sum_{j \neq \ell, k} E(q_\ell q_k \boldsymbol{\varepsilon}'_\ell \mathbf{M}_\tau \mathbf{s}_{j,-1} \mathbf{s}'_{j,-1} \mathbf{M}_\tau \boldsymbol{\varepsilon}_k)$$

[A.14]

$$\begin{aligned}
&= \frac{T^2}{N^2} \sum_{j \neq \ell, k} E \left(q_\ell q_k \boldsymbol{\varepsilon}'_\ell \mathbf{M}_\tau \mathbf{H} \boldsymbol{\varepsilon}_{j,-1} \boldsymbol{\varepsilon}'_{j,-1} \mathbf{H}' \mathbf{M}_\tau \boldsymbol{\varepsilon}_k \right) \\
&= \frac{T^2}{N^2} \sum_{j \neq \ell, k} E \left[q_\ell q_k \boldsymbol{\varepsilon}'_\ell \mathbf{M}_\tau \mathbf{H} E \left(\boldsymbol{\varepsilon}_{j,-1} \boldsymbol{\varepsilon}'_{j,-1} \right) \mathbf{H}' \mathbf{M}_\tau \boldsymbol{\varepsilon}_k \right] \\
&= \frac{T^2(N-2)}{N^2} \left(\frac{1}{N-2} \sum_{j \neq \ell, k} \sigma_j^2 \right) E \left[q_\ell q_k \boldsymbol{\varepsilon}'_\ell \mathbf{M}_\tau \mathbf{H} \mathbf{H}' \mathbf{M}_\tau \boldsymbol{\varepsilon}_k \right] = O \left(N^{-1} T^{-1/2} \right).
\end{aligned}$$

The last equality follows since $E \left(q_\ell q_k \boldsymbol{\varepsilon}'_\ell \mathbf{M}_\tau \mathbf{H} \mathbf{H}' \mathbf{M}_\tau \boldsymbol{\varepsilon}_k \right) = O \left(T^{-5/2} \right)$. Therefore, double summing $O \left(N^{-1} T^{-1/2} \right)$ term across cross sectional units and dividing it by N gives $O \left(T^{-1/2} \right)$, we have

$$Var \left(N^{-1/2} \sum_{i=1}^N J_{2i} \right) = O \left(N^{-1} \right) + O \left(T^{-1/2} \right). \quad \blacksquare$$

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